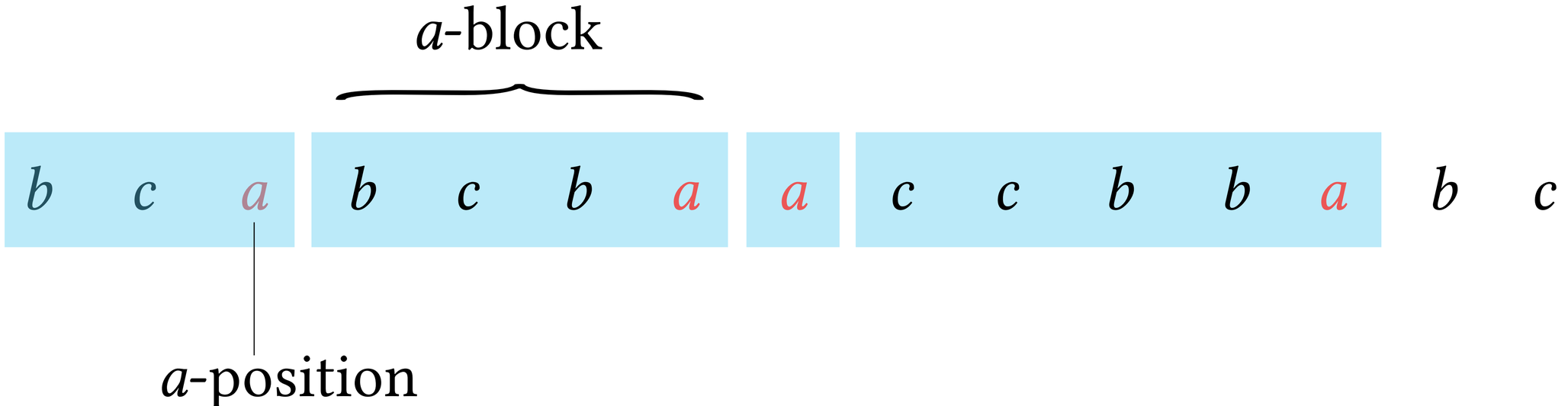
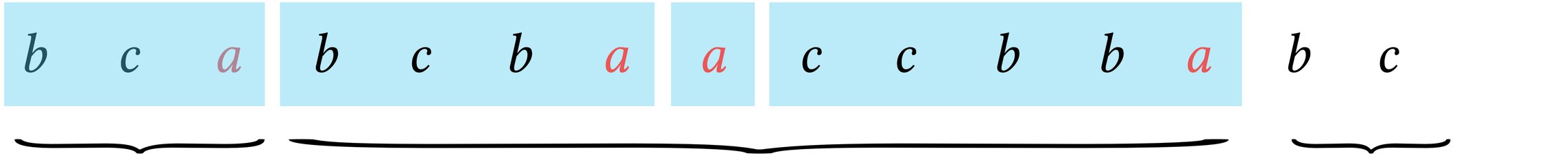
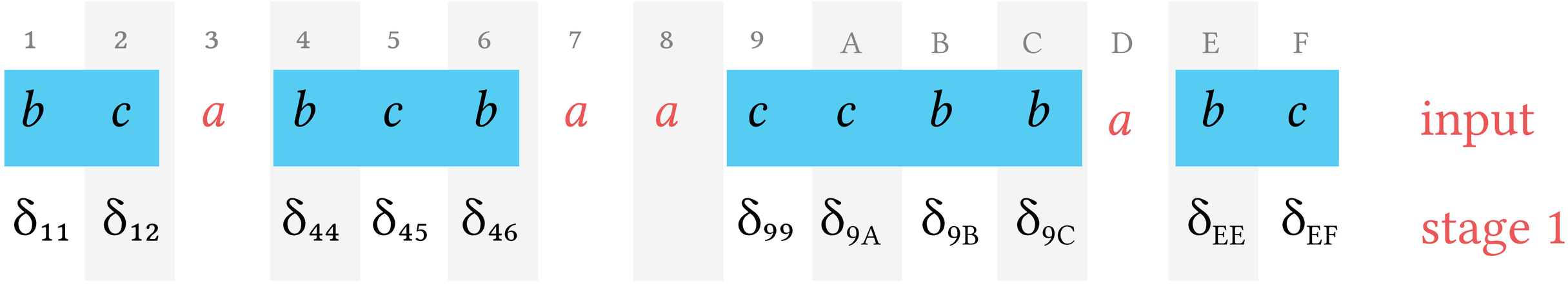
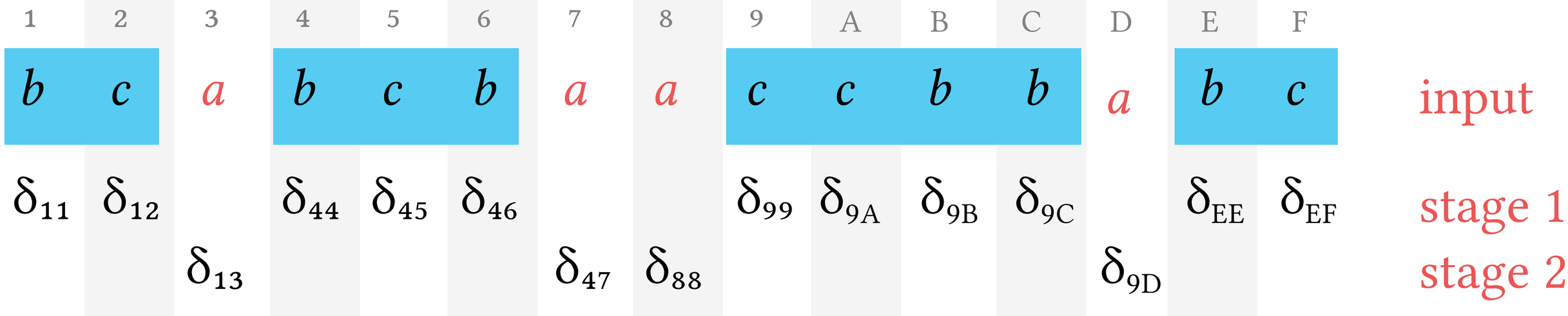
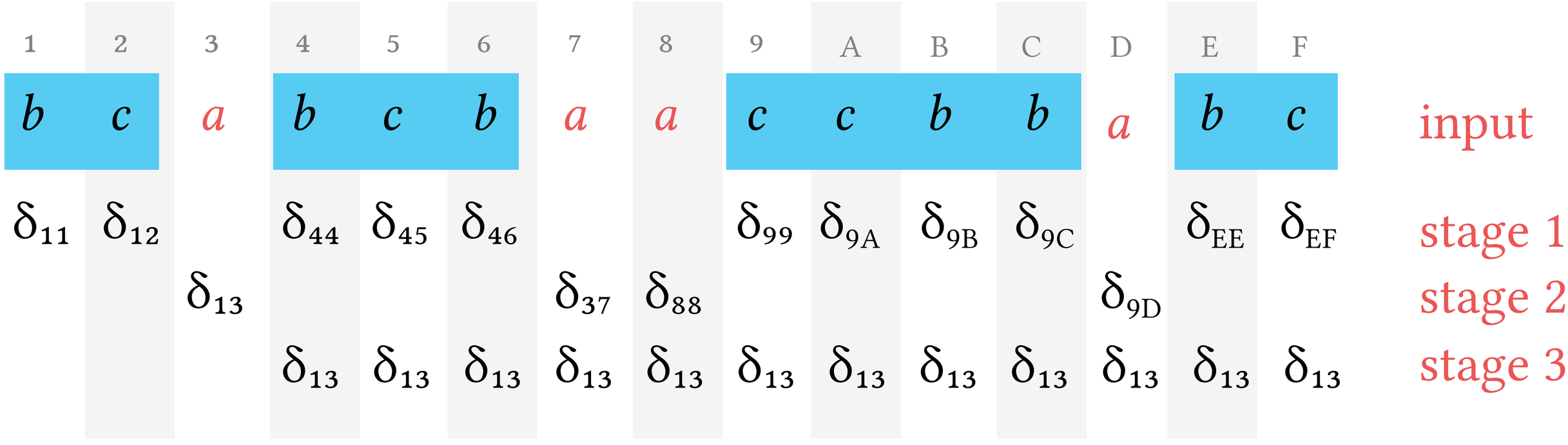
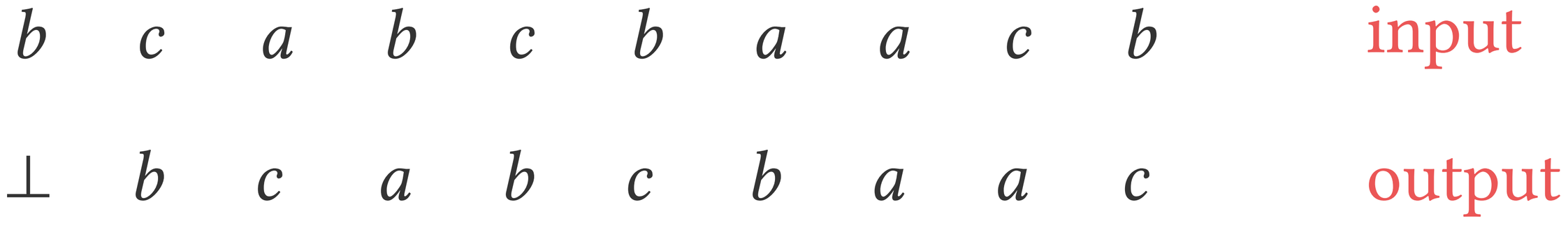

a-block
b c a b c b a a c c b b a b c
a-position

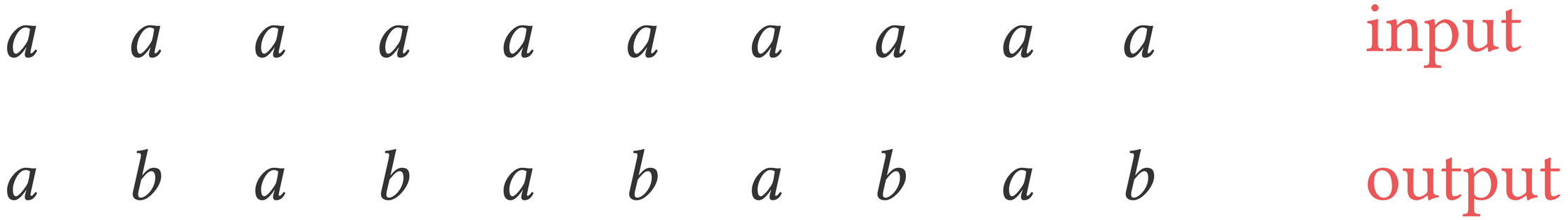

b c a b c b a a c c b b a b c
first part:
first a-block
middle part:
all a-blocks except the first
last part:
a-free suffix

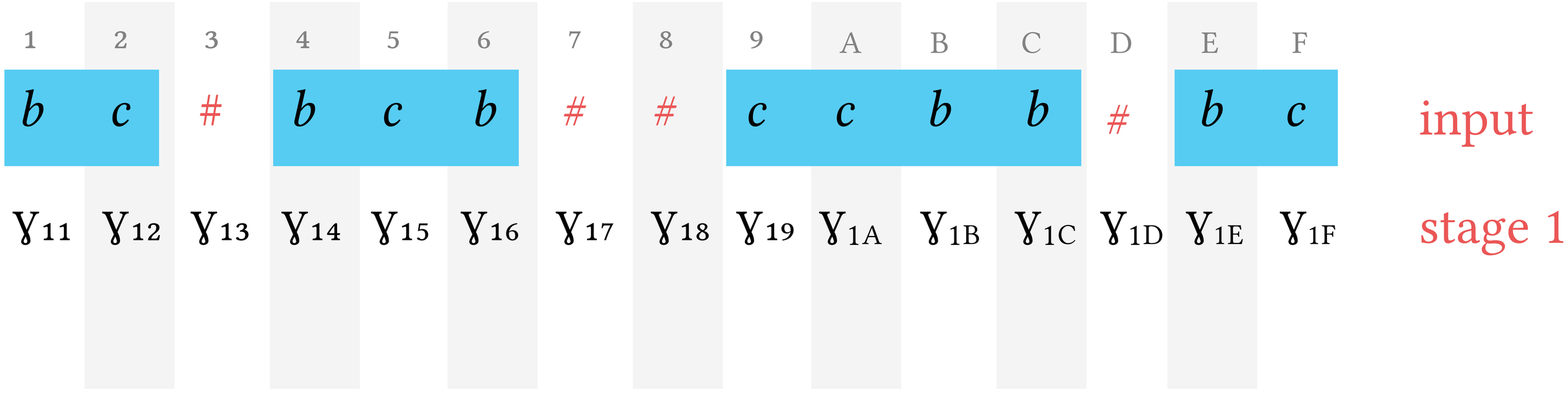

1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1

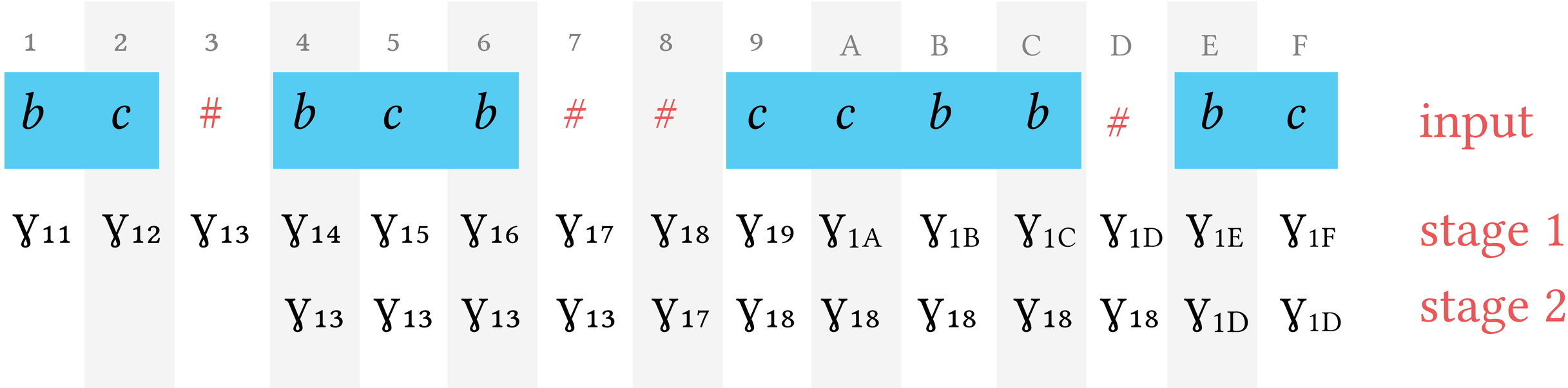

1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1
δ13 δ47 δ88 δ9D
stage 2

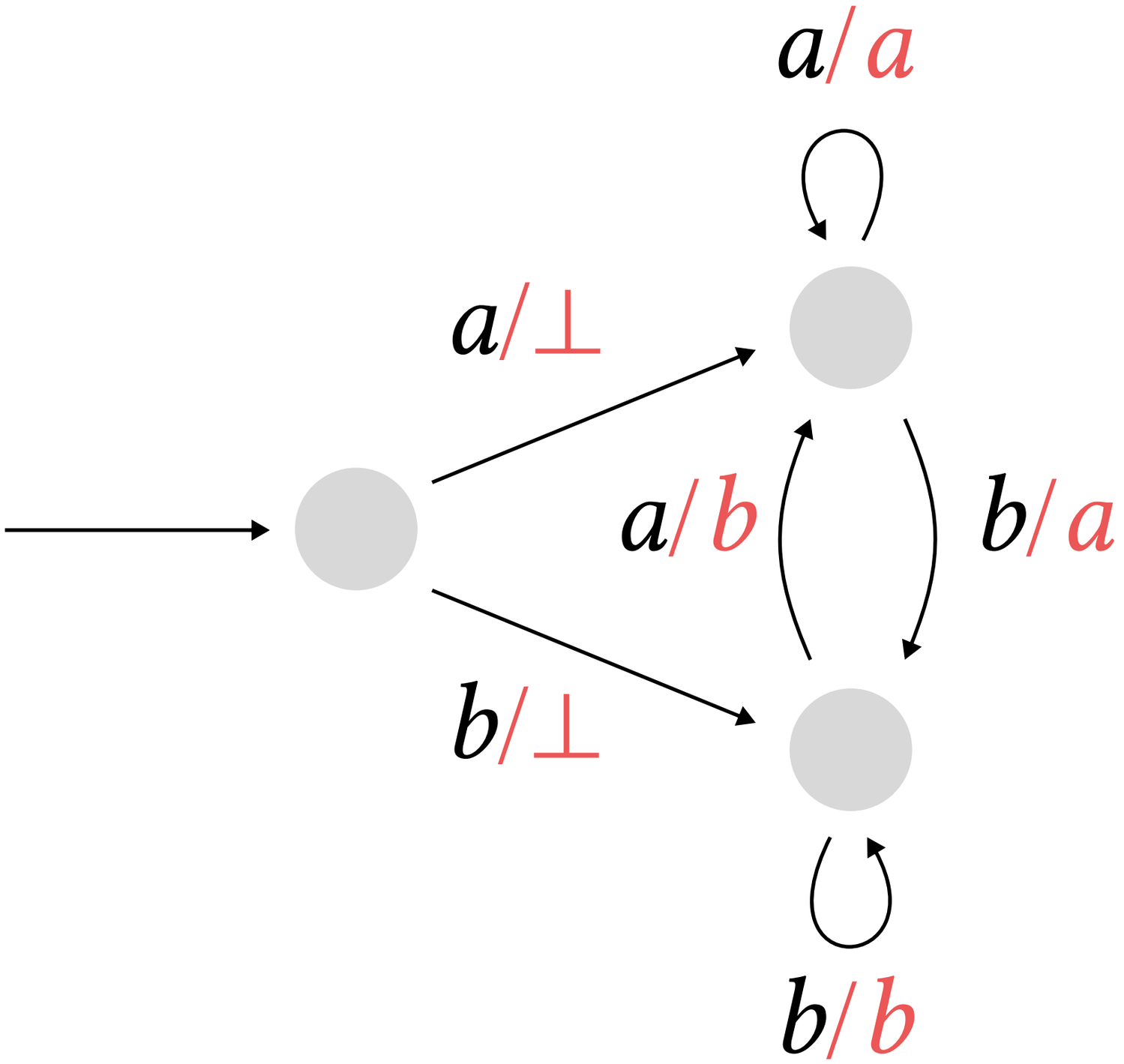

1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1
δ13 δ37 δ88 δ9D
stage 2
δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13
stage 3

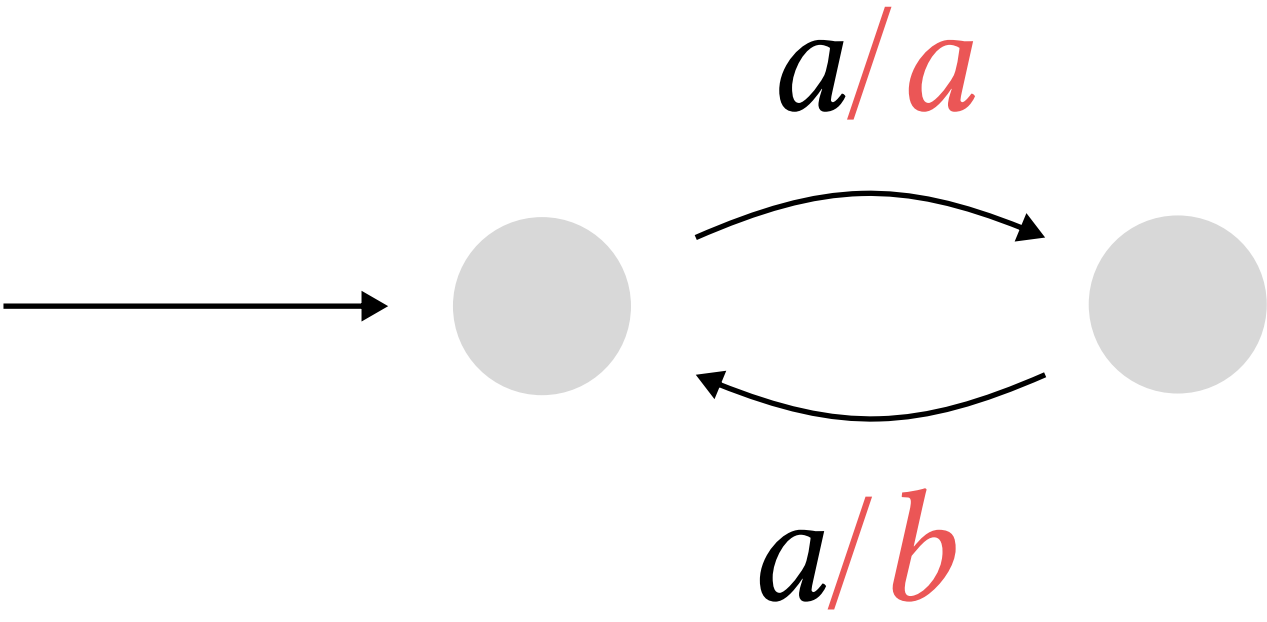

b c a b c b a a c b
input
⊥ b c a b c b a a c
output

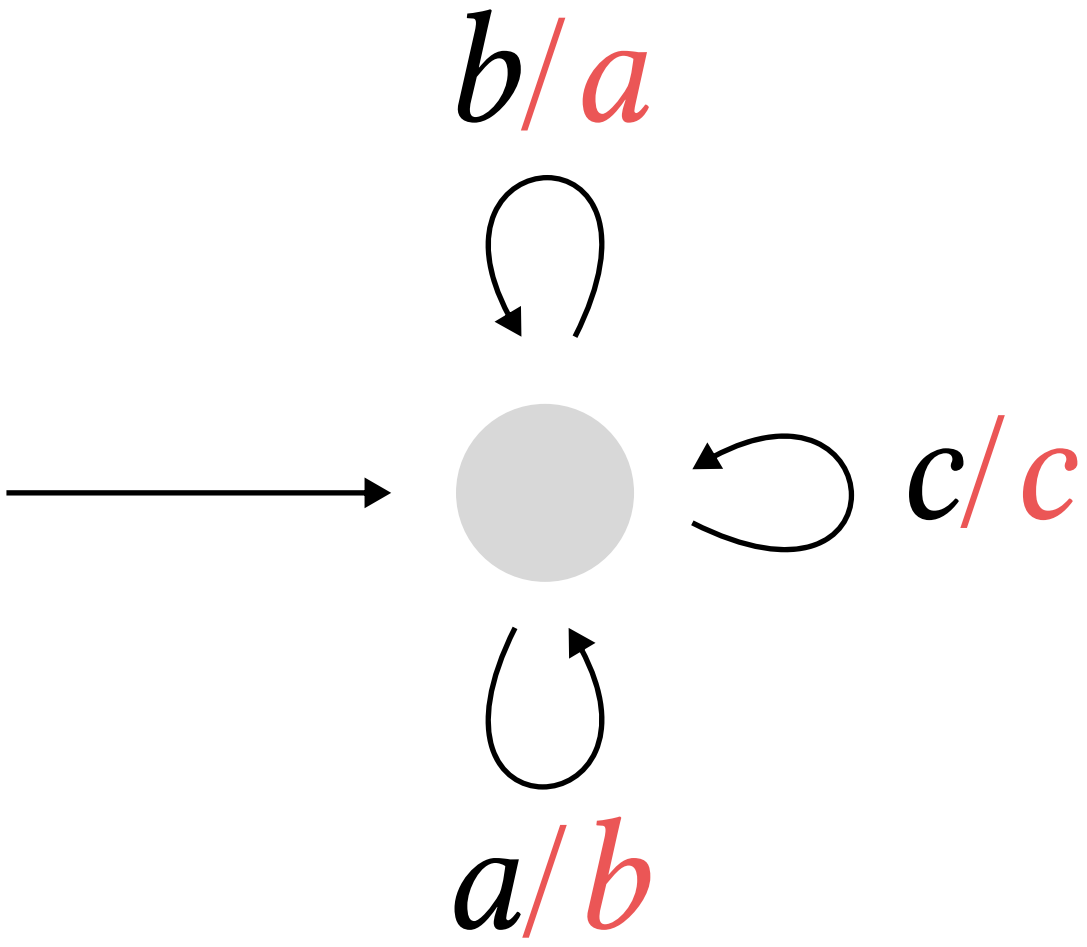

a a a a a a a a a a
input
a b a b a b a b a b
output


1 2 3 4 5 6 7 8 9 A B C D E F
b c # b c b # # c c b b # b c
input
Y11 Y12 Y13 Y14 Y15 Y16 Y17 Y18 Y19 Y1A Y1B Y1C Y1D Y1E Y1F
stage 1


1 2 3 4 5 6 7 8 9 A B C D E F
b c # b c b # # c c b b # b c
input
Y_{11} Y_{12} Y_{13} Y_{14} Y_{15} Y_{16} Y_{17} Y_{18} Y_{19} Y_{1A} Y_{1B} Y_{1C} Y_{1D} Y_{1E} Y_{1F}
stage 1
Y_{13} Y_{13} Y_{13} Y_{13} Y_{17} Y_{18} Y_{18} Y_{18} Y_{18} Y_{18} Y_{1D} Y_{1D}
stage 2


a/a
a/⊥
a/b
b/a
b/⊥
b/b


a/a
a/b


b/a
c/c
a/b

a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 a_9

q_0 q_1 q_2 q_3 q_4 q_5 q_6 q_7 q_8 q_9 run of f

b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8 b_9

p_0 p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8 p_9 run of g

c_1 c_2 c_3 c_4 c_5 c_6 c_7 c_8 c_9

input letter

a

source state

q

b

p

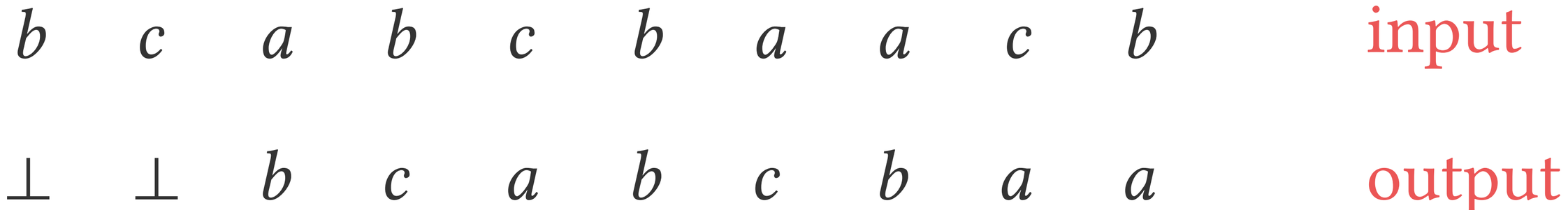
q’

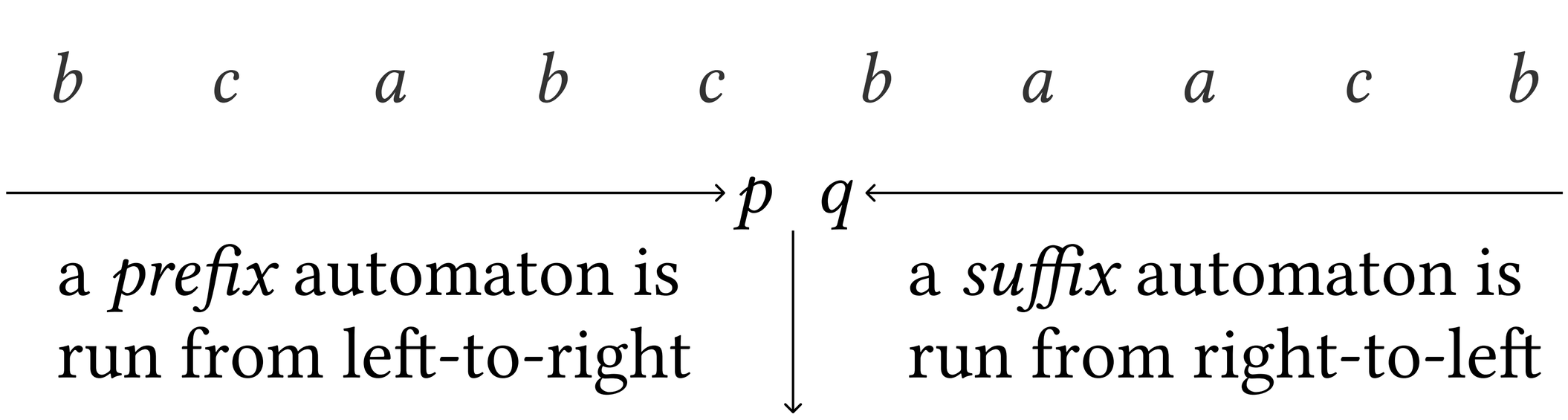
p’

target state

c

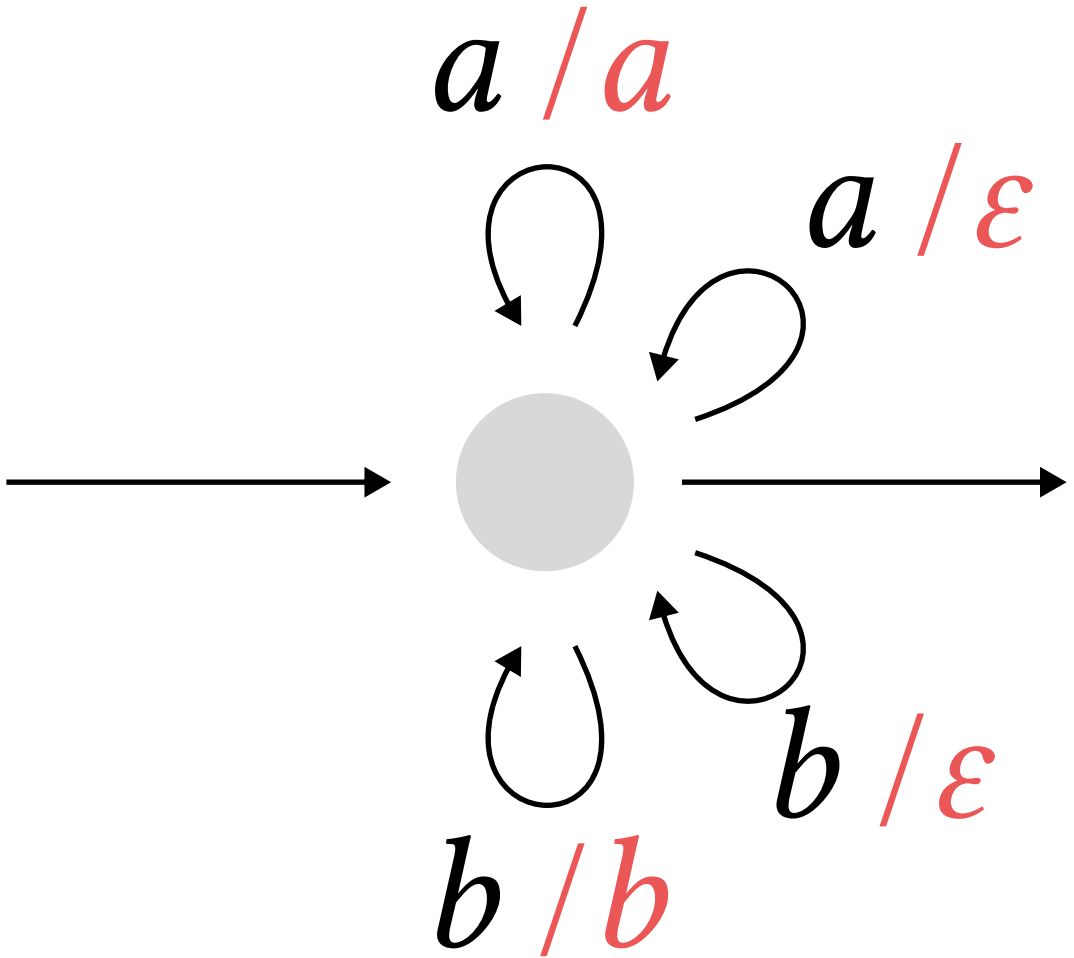
output letter

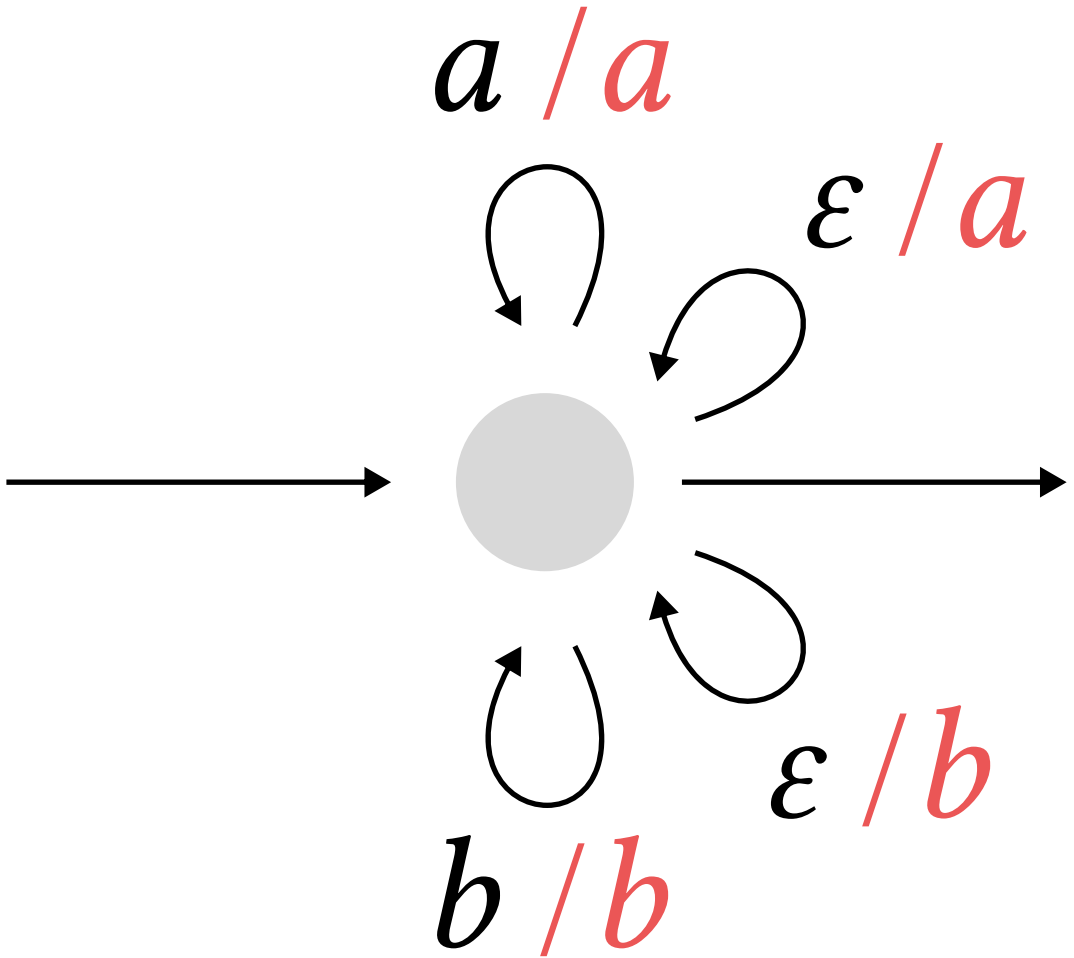

b c a b c b a a c b
input
⊥ ⊥ b c a b c b a a
output



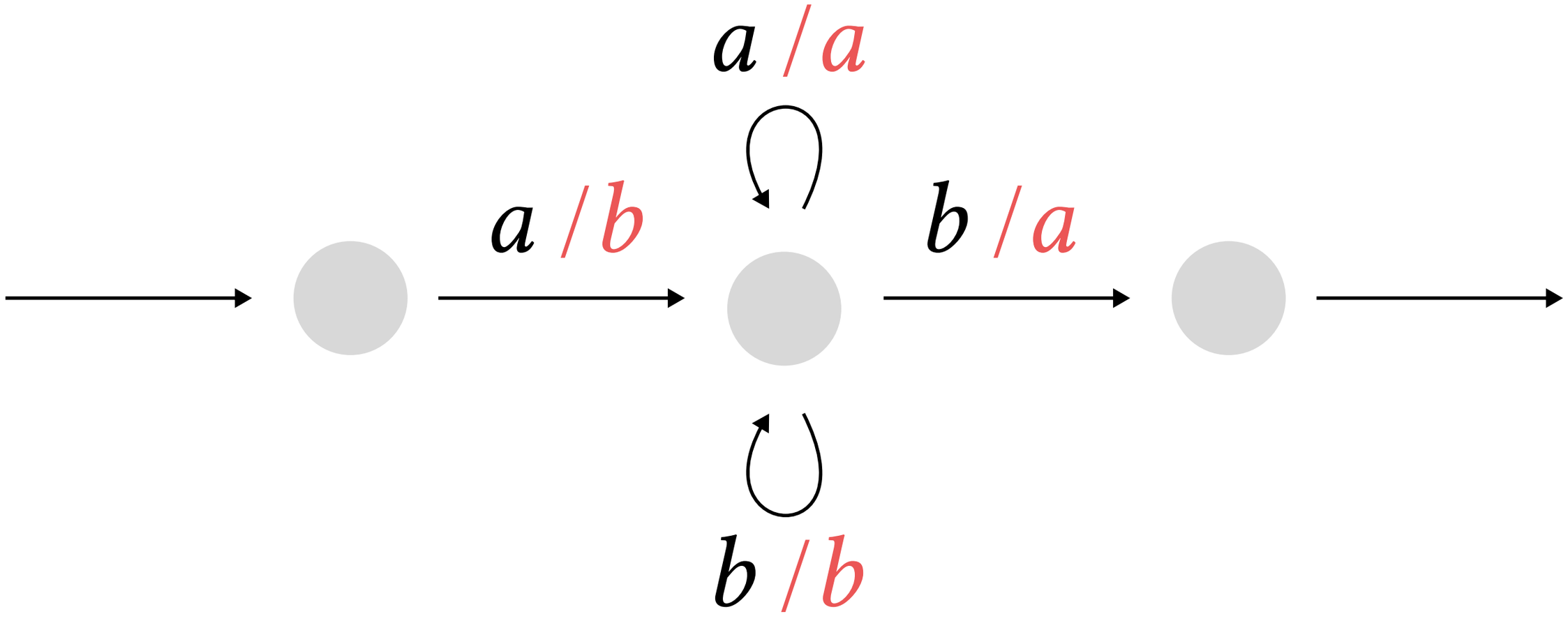
cdd

based on the states of the prefix and suffix automata, a piece of the output is produced


a / a
a / ε
b / ε
b / b


a / a
ε / a
ε / b
b / b

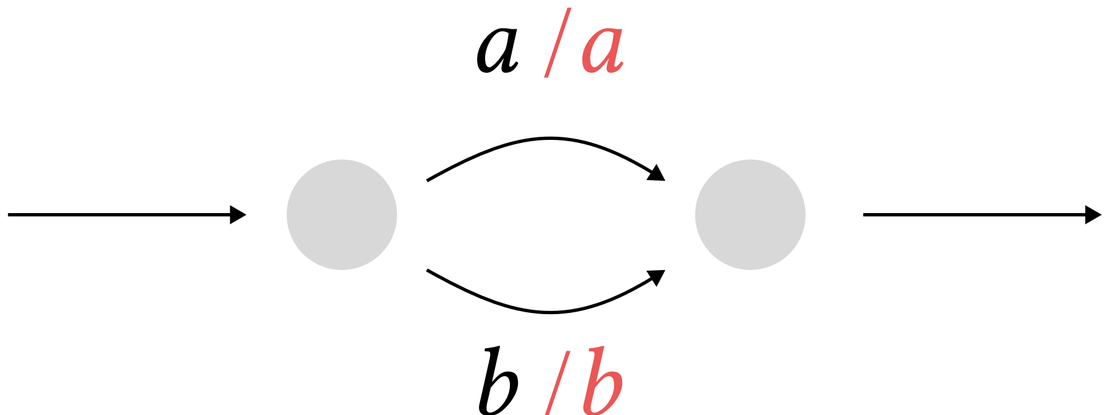
inputs that start with f and end with b

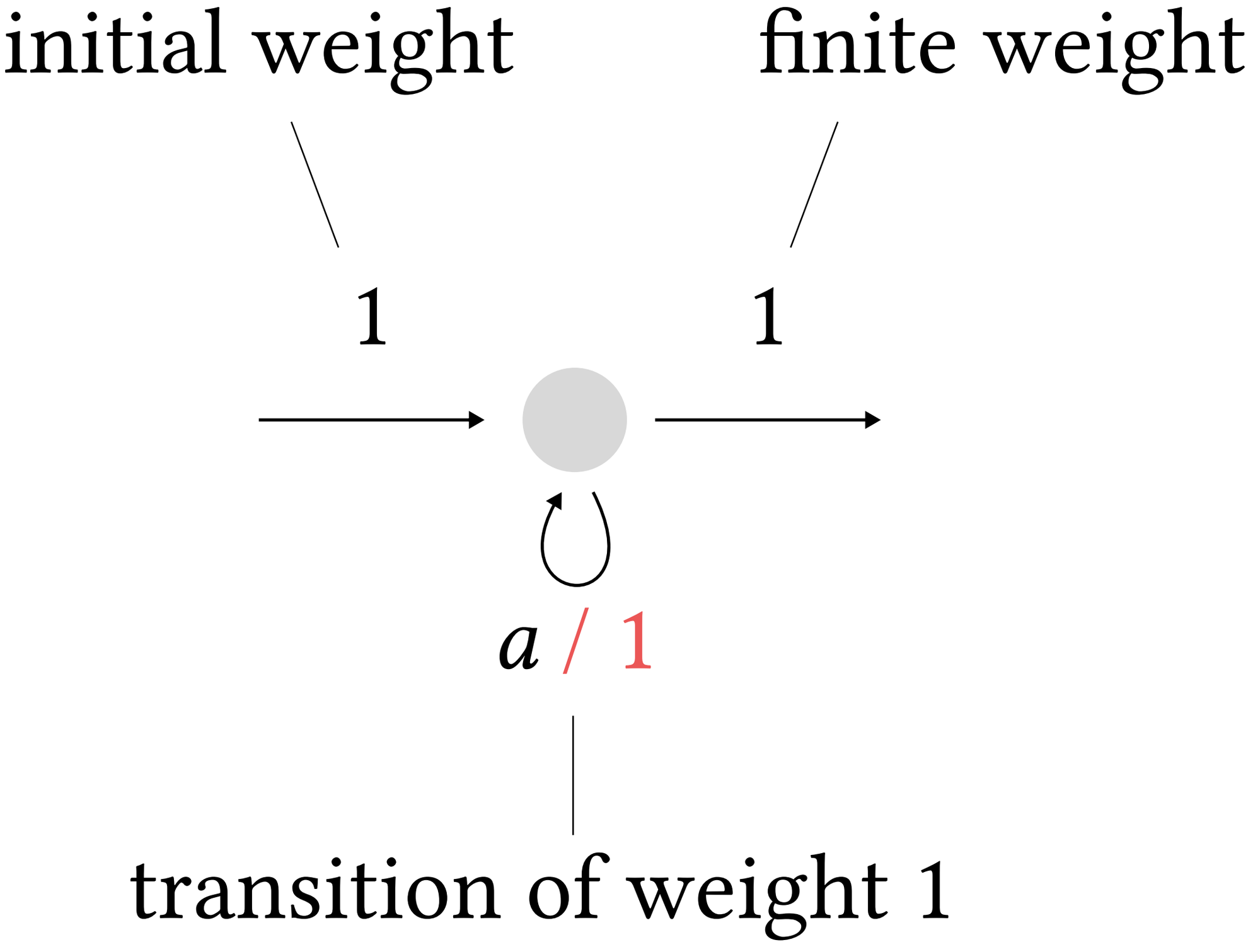


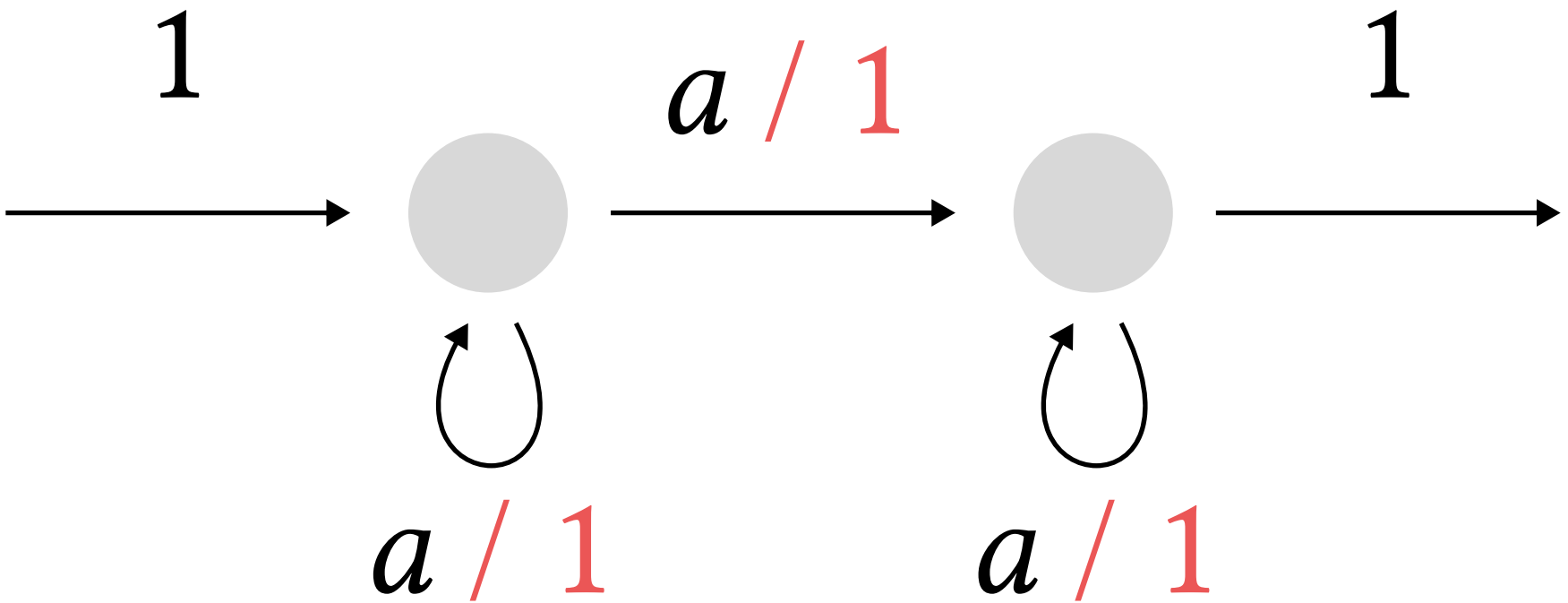
empty inputs

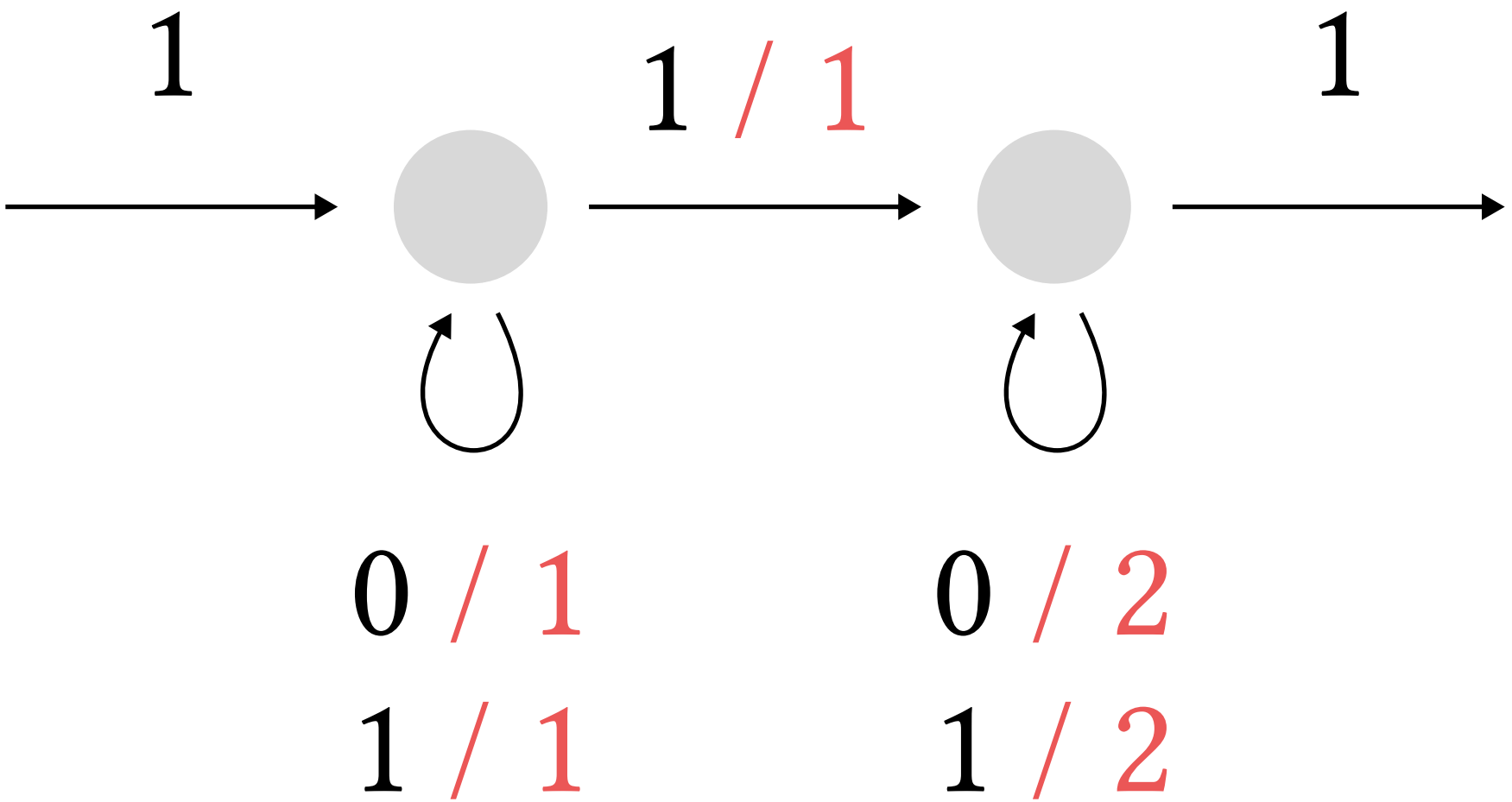


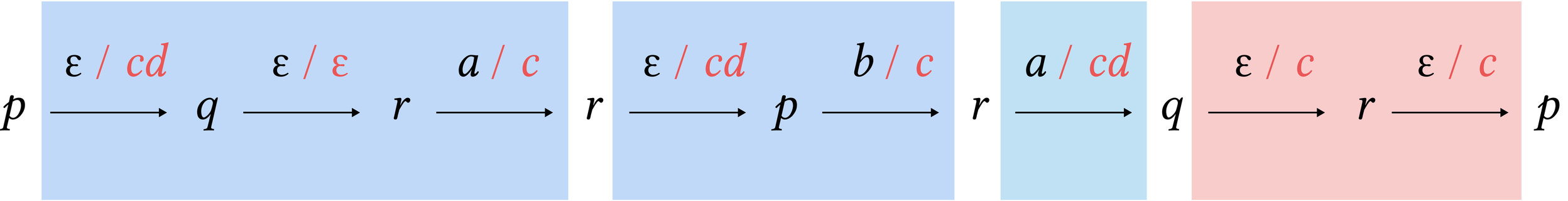
one letter inputs

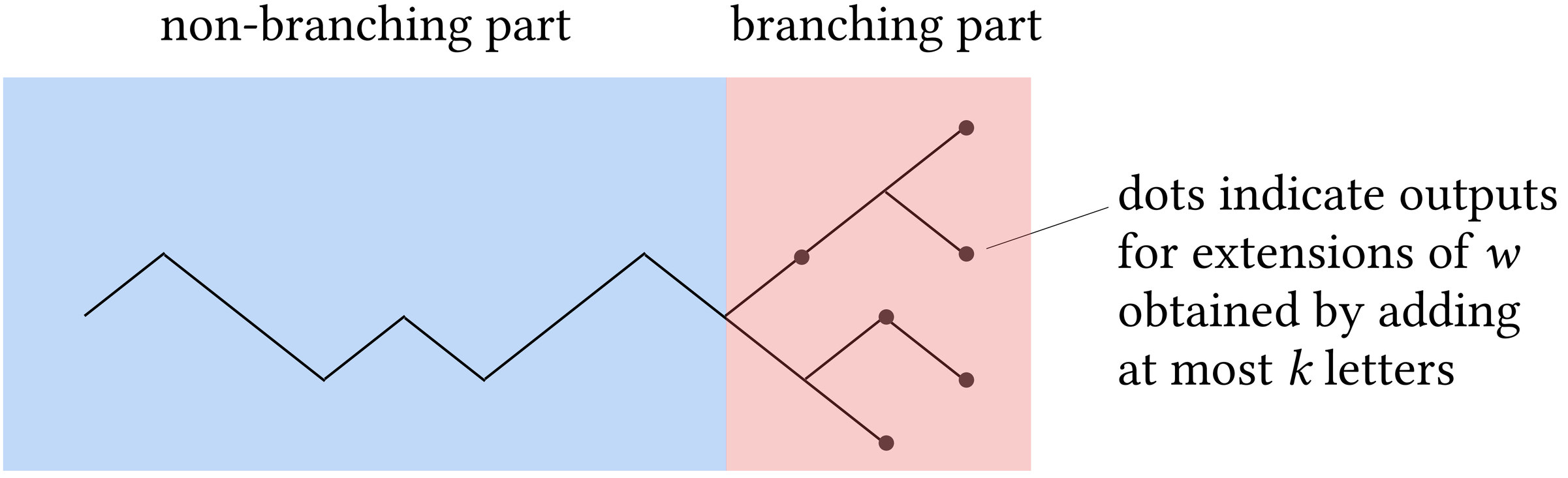


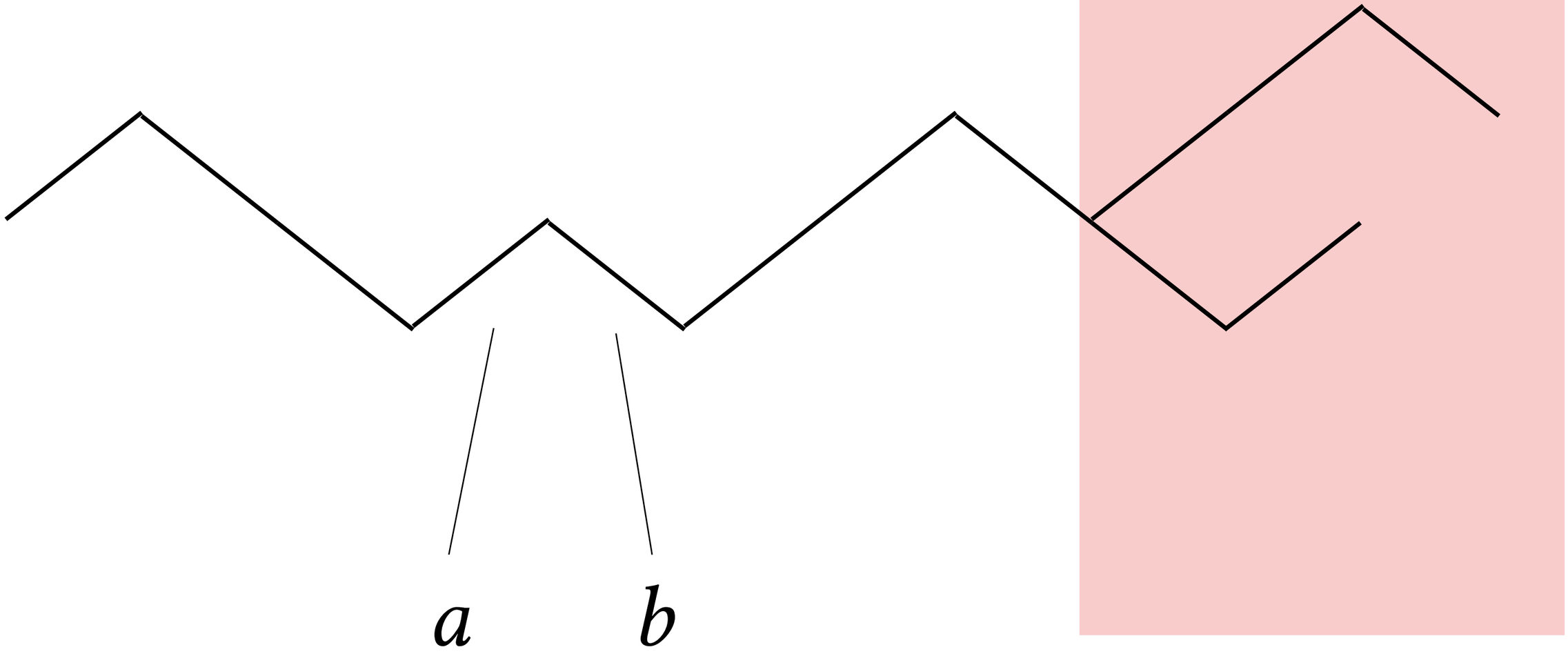

initial weight
finite weight
1
1
a / 1
transition of weight 1


1
a / 1
1
a / 1
a / 1


1
1 / 1
1
0 / 1
1 / 1
0 / 2
1 / 2

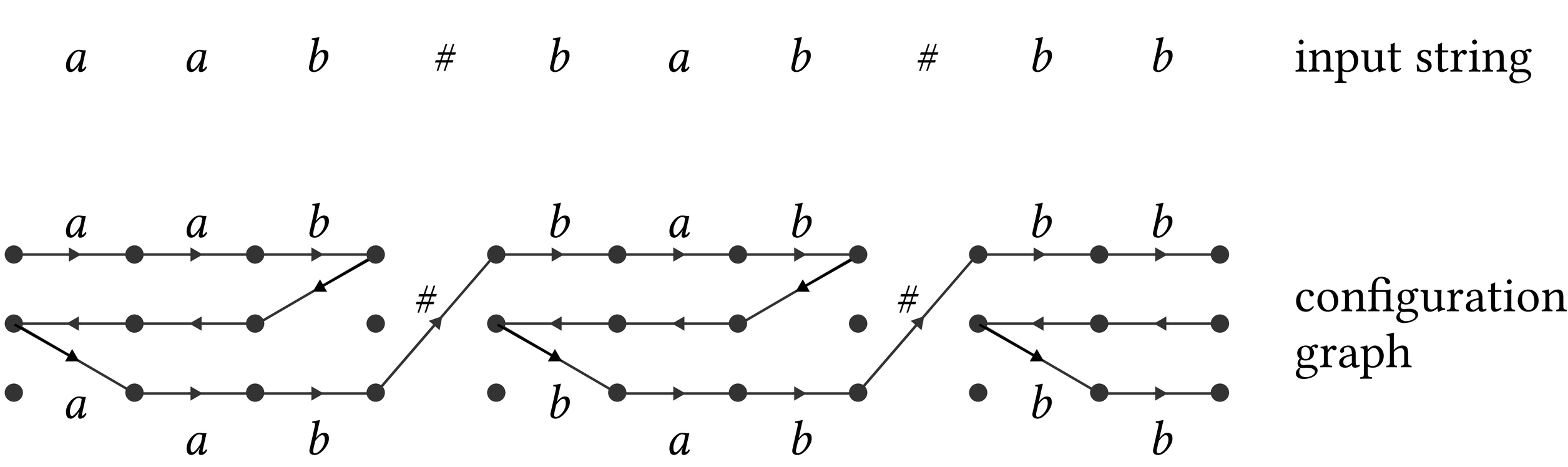

ε / cd
ε / ε
a / c
ε / cd
b / c
a / cd
ε / c
ε / c
p
q
r
r
p
r
q
r
p

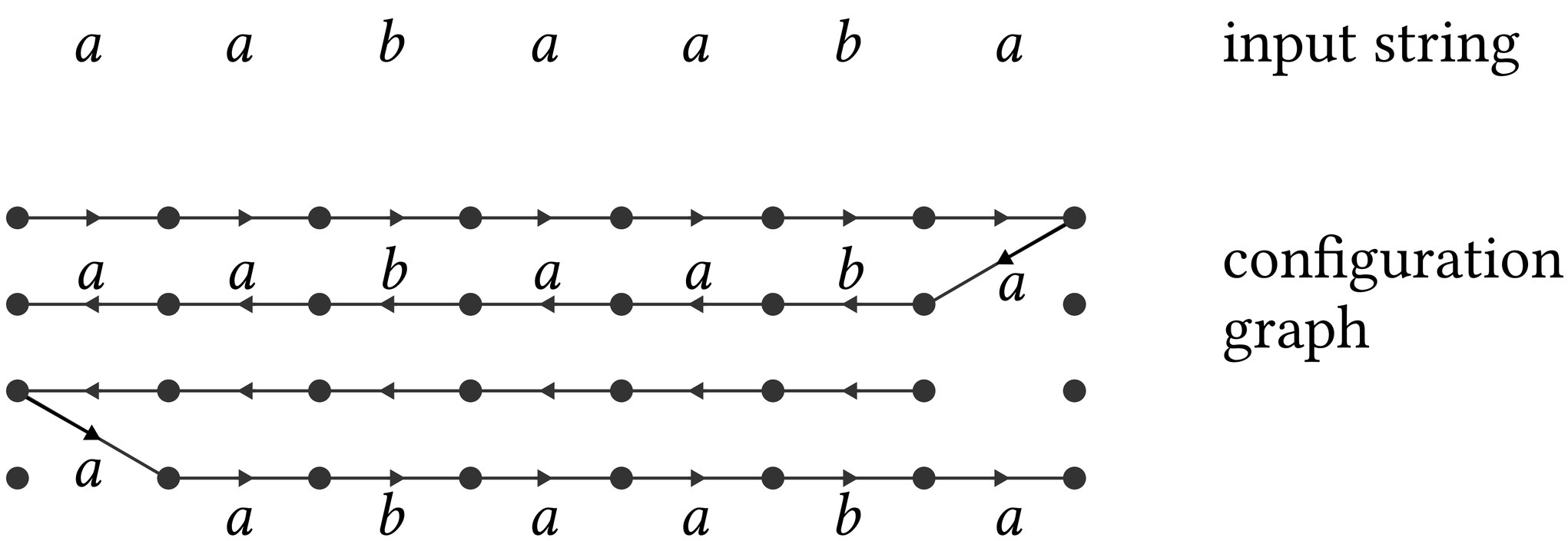

non-branching part
branching part
dots indicate outputs
for extensions of w
obtained by adding
at most k letters

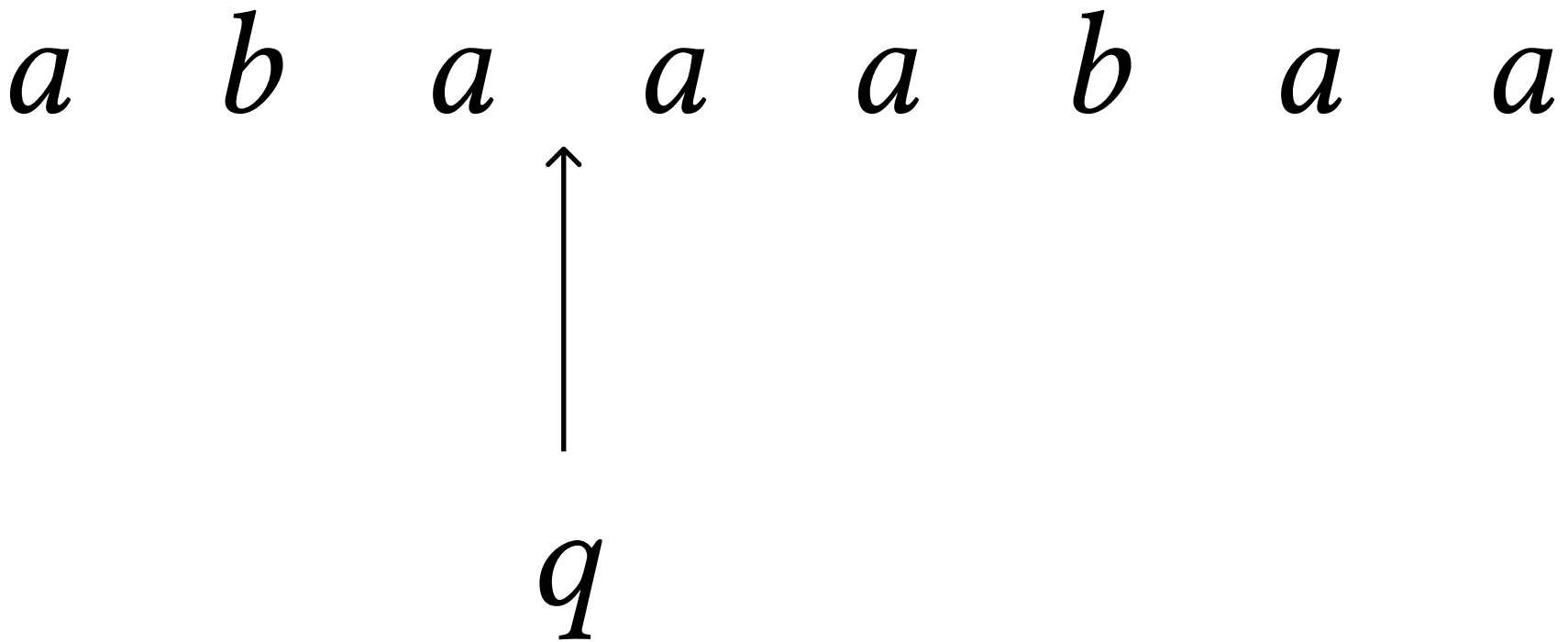


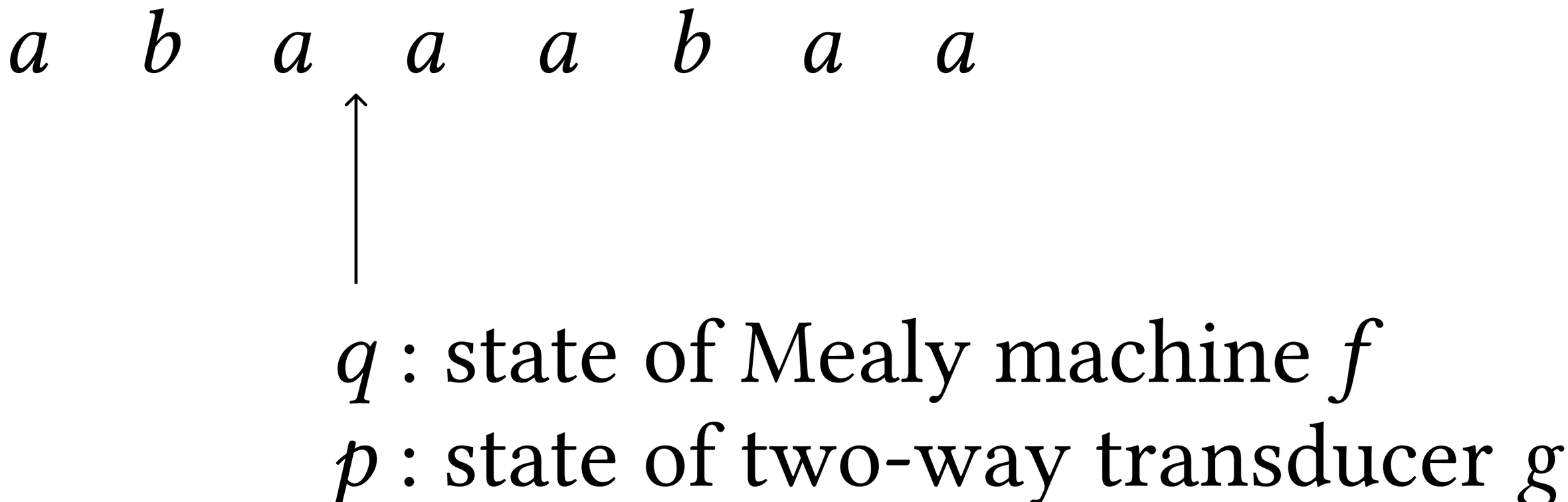
directions indicate letters

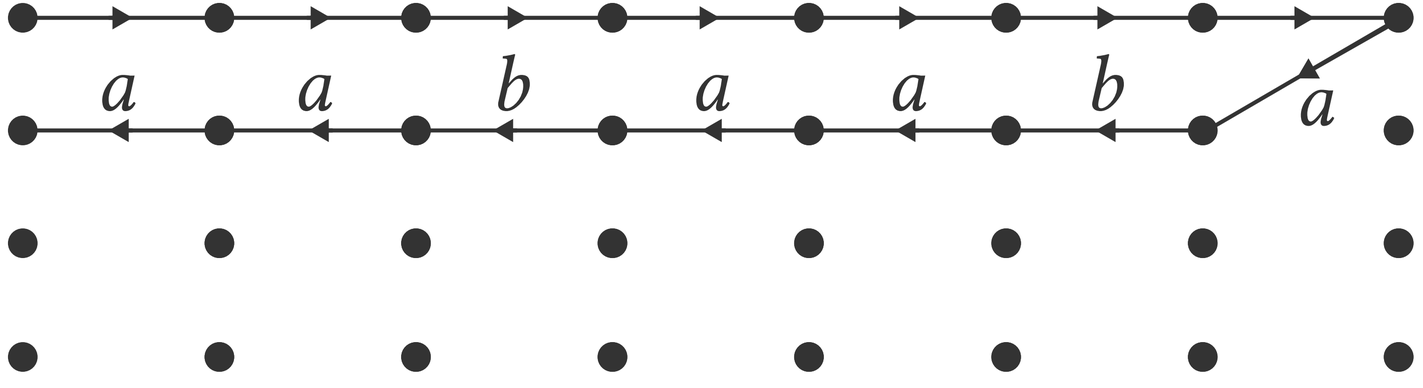
left distance $k = 3$


a a b # b a b # b b
input string
a a b # b a b # b b
a a b b a b b b
configuration graph

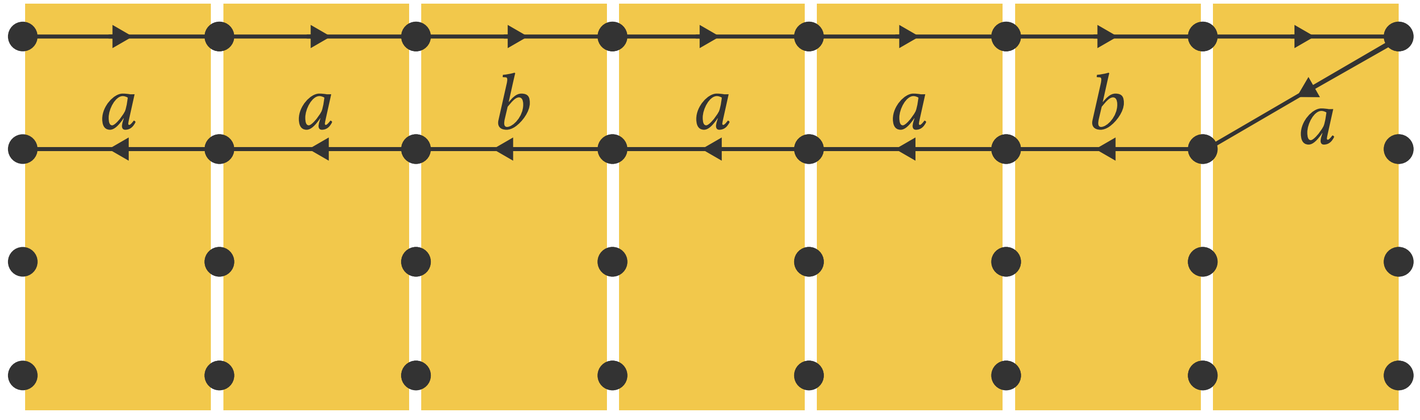

a a b a a b a
input string
a a b a a b a
a a b a a b a
configuration
graph


a b a a a b a a
q


a b a a a b a a
q : state of Mealy machine f
p : state of two-way transducer g

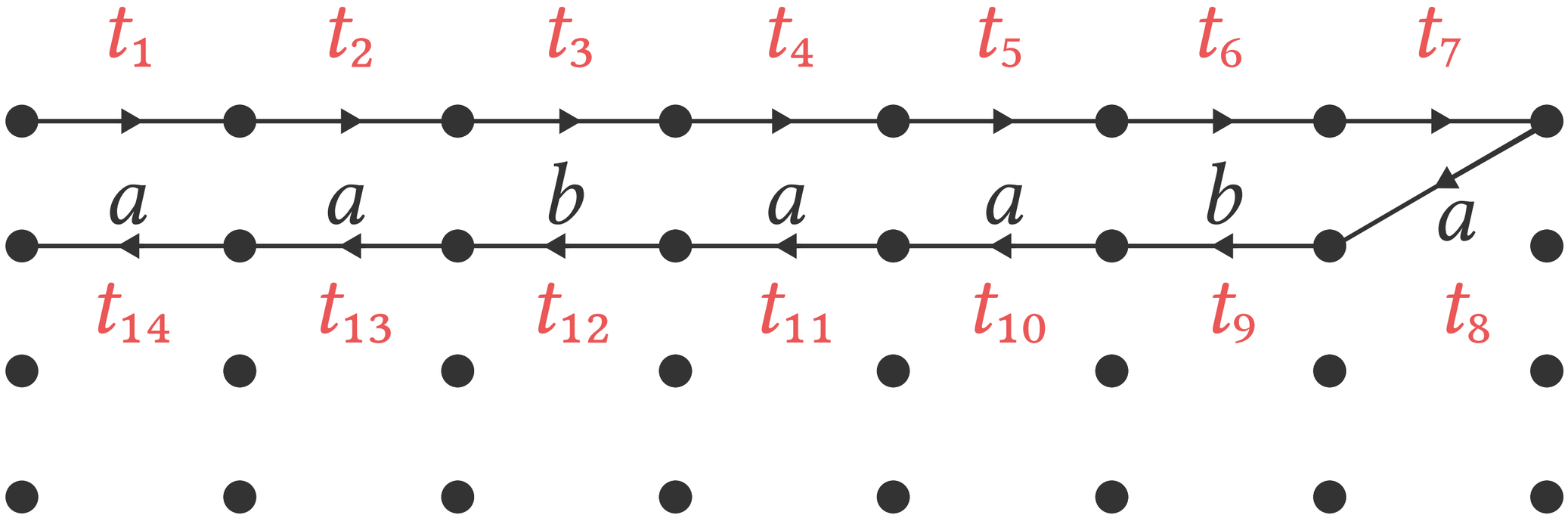


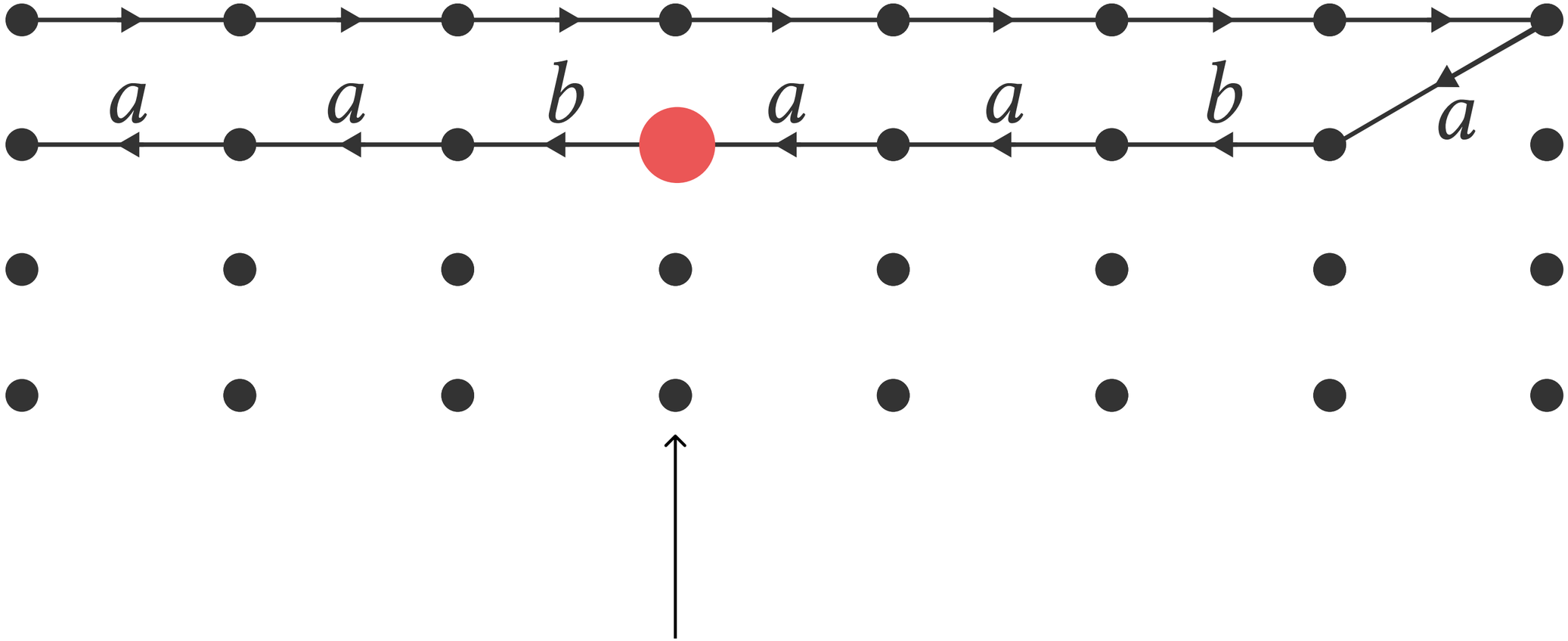
reachable part of the configuration graph



string
representation

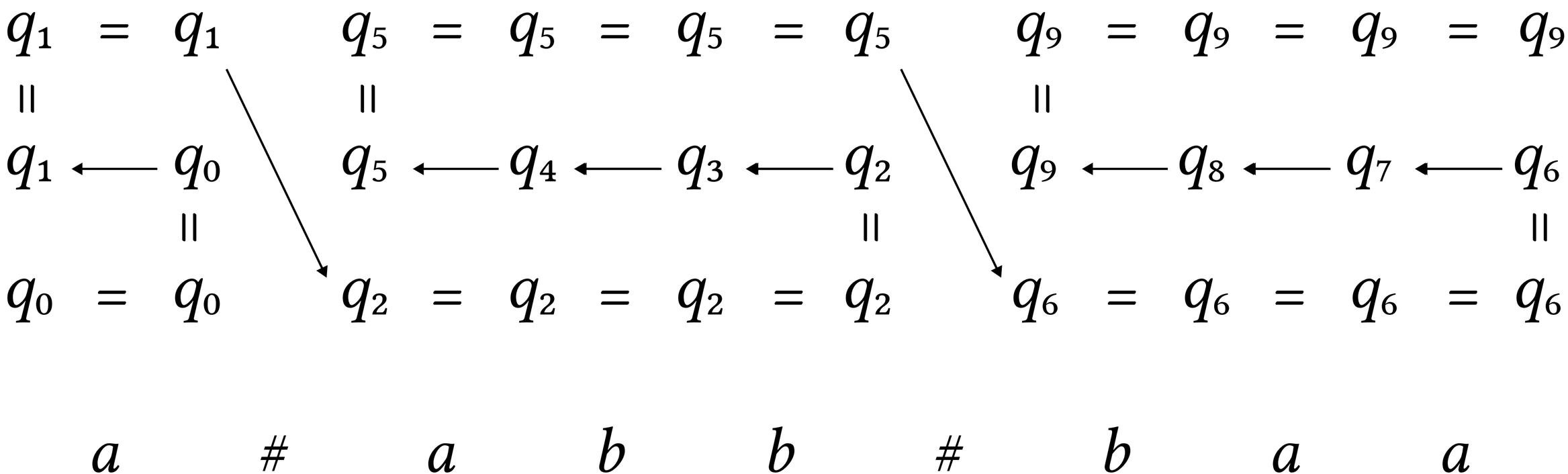


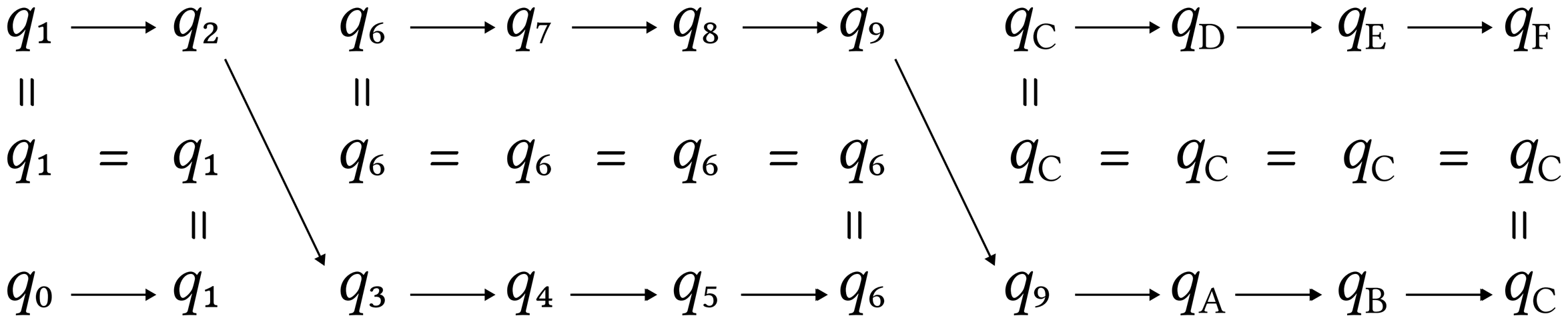

t_1 t_2 t_3 t_4 t_5 t_6 t_7
a a b a a b a
t_{14} t_{13} t_{12} t_{11} t_{10} t_9 t_8

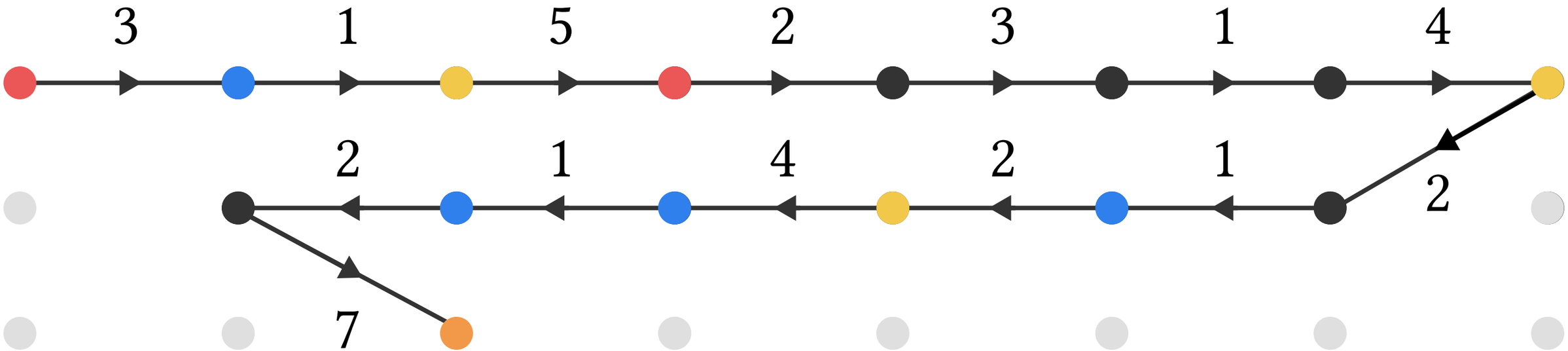


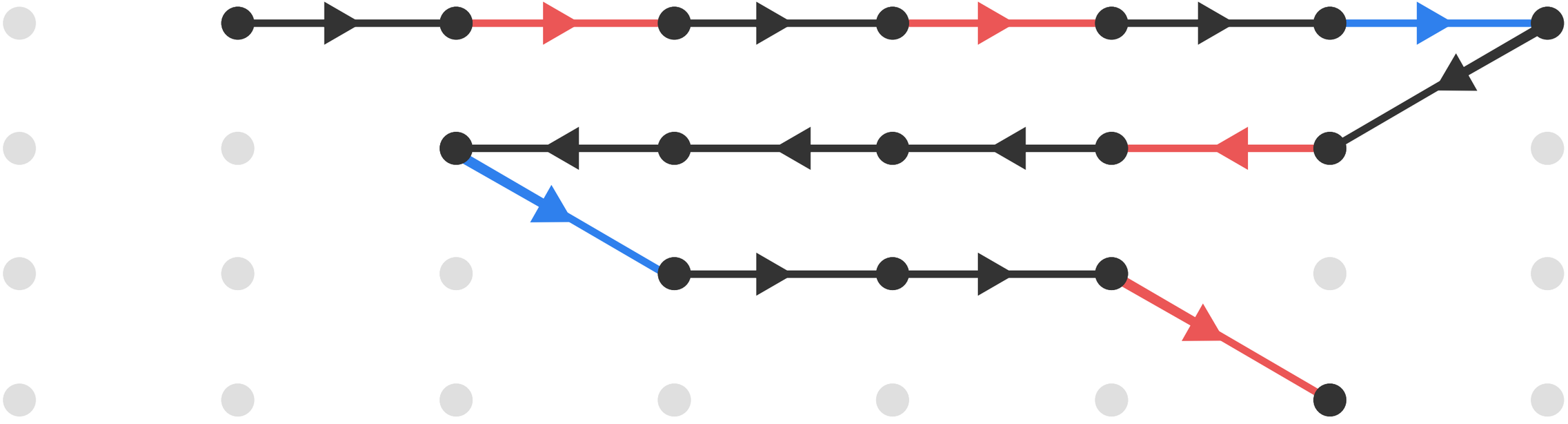
product automaton remembers:

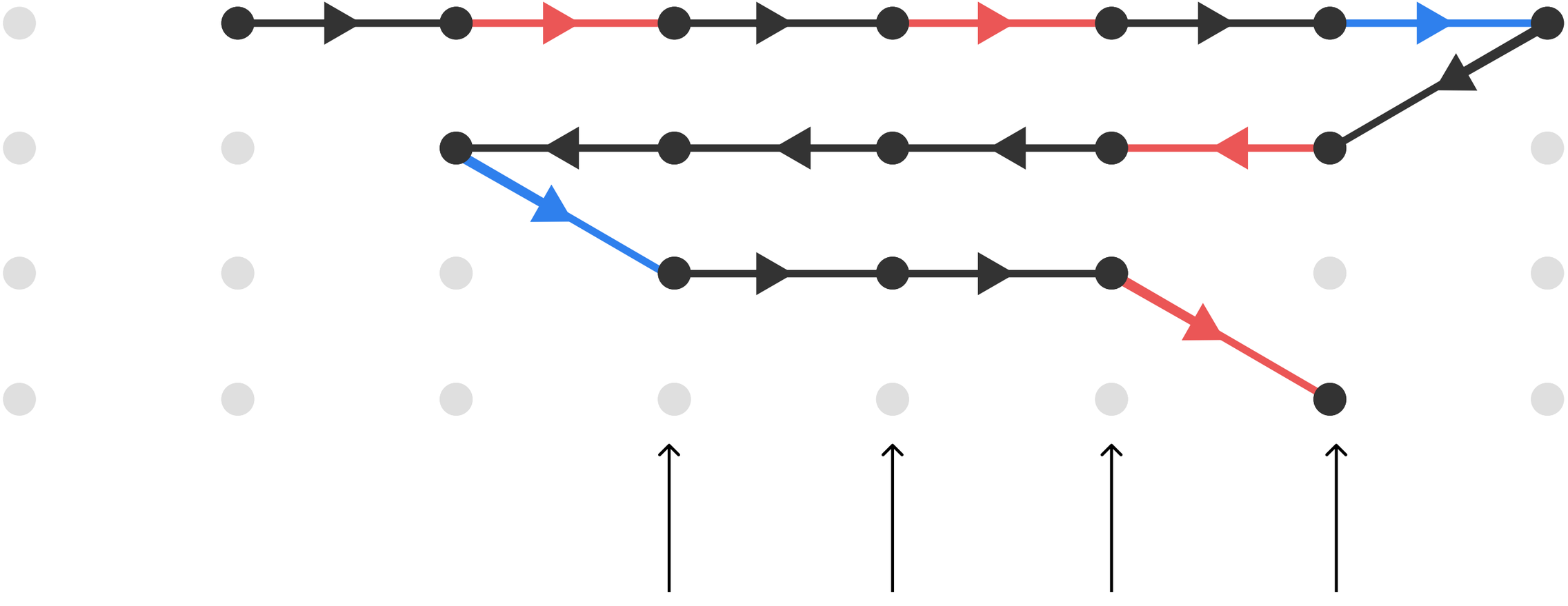
- state of f (red circle); and
- state of g

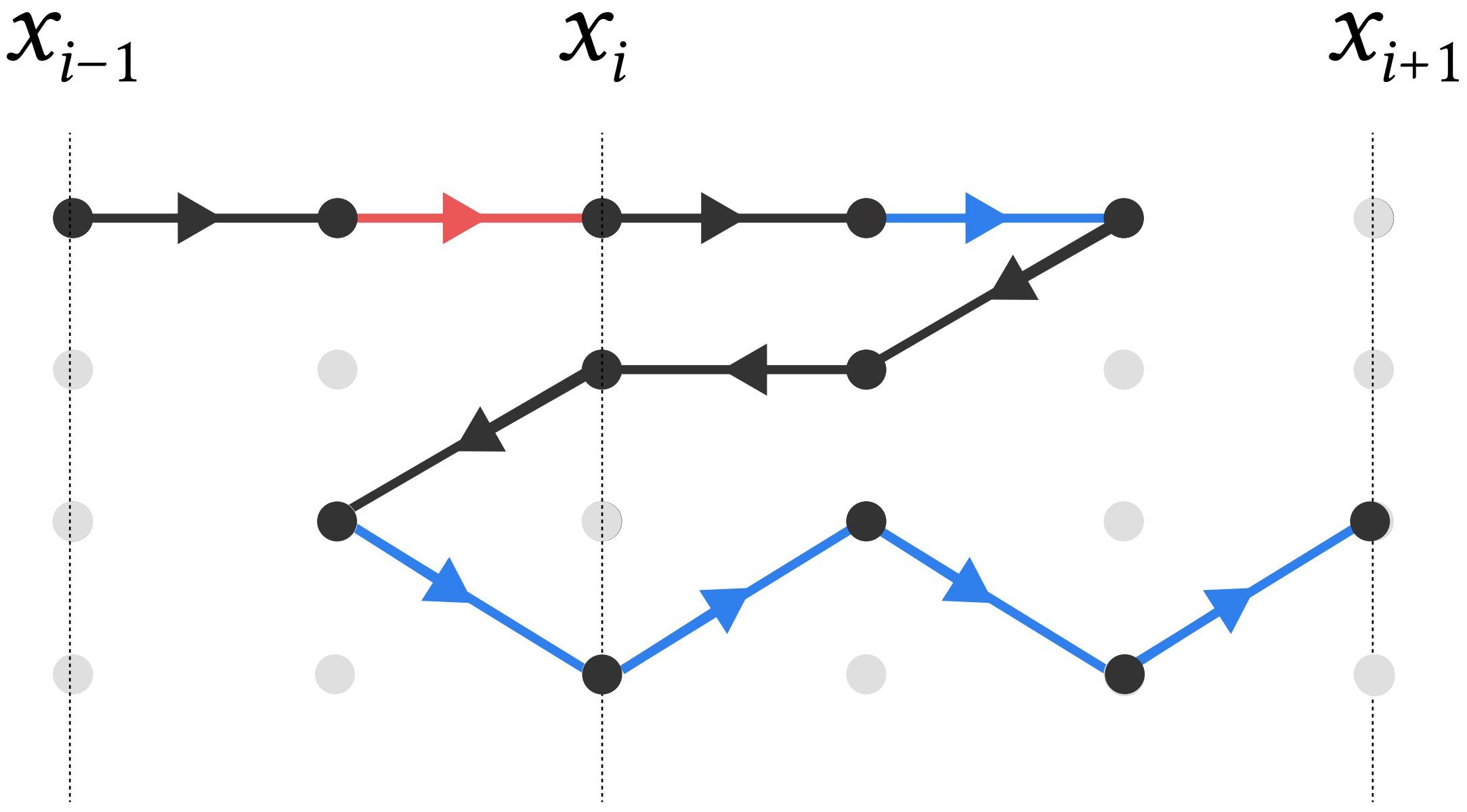

$q_1 = q_1$ $q_5 = q_5 = q_5 = q_5$ $q_9 = q_9 = q_9 = q_9$
$q_1 \leftarrow q_0$ $q_5 \leftarrow q_4 \leftarrow q_3 \leftarrow q_2$ $q_9 \leftarrow q_8 \leftarrow q_7 \leftarrow q_6$
$q_0 = q_0$ $q_2 = q_2 = q_2 = q_2$ $q_6 = q_6 = q_6 = q_6$
a # a b b # b a a

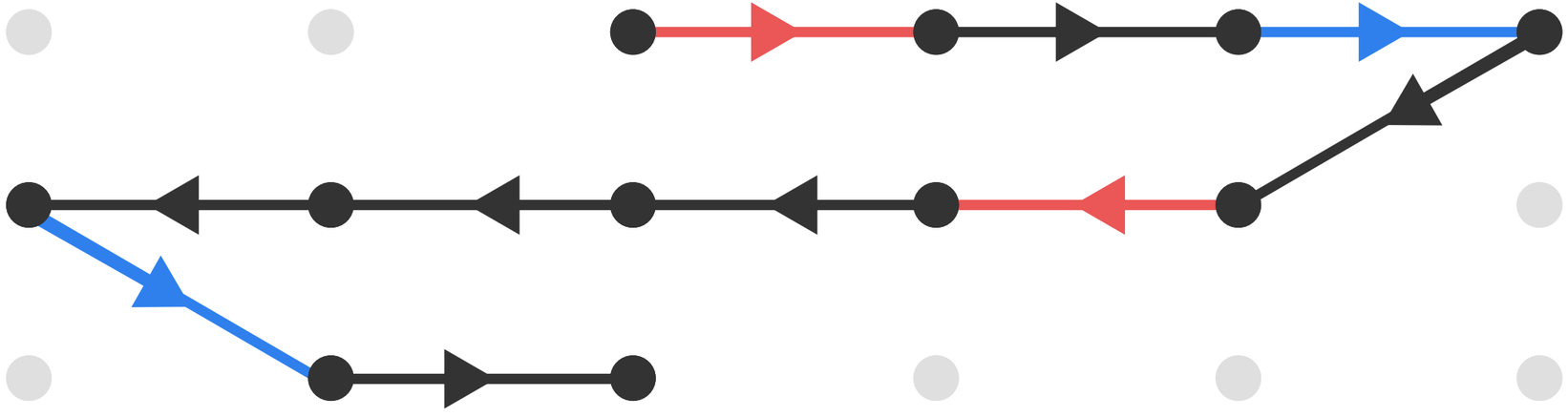

$q_1 \longrightarrow q_2$
$q_6 \longrightarrow q_7 \longrightarrow q_8 \longrightarrow q_9$
$q_C \longrightarrow q_D \longrightarrow q_E \longrightarrow q_F$
$q_1 = q_1$
$q_6 = q_6 = q_6 = q_6$
$q_C = q_C = q_C = q_C$
$q_0 \longrightarrow q_1$
$q_3 \longrightarrow q_4 \longrightarrow q_5 \longrightarrow q_6$
$q_9 \longrightarrow q_A \longrightarrow q_B \longrightarrow q_C$

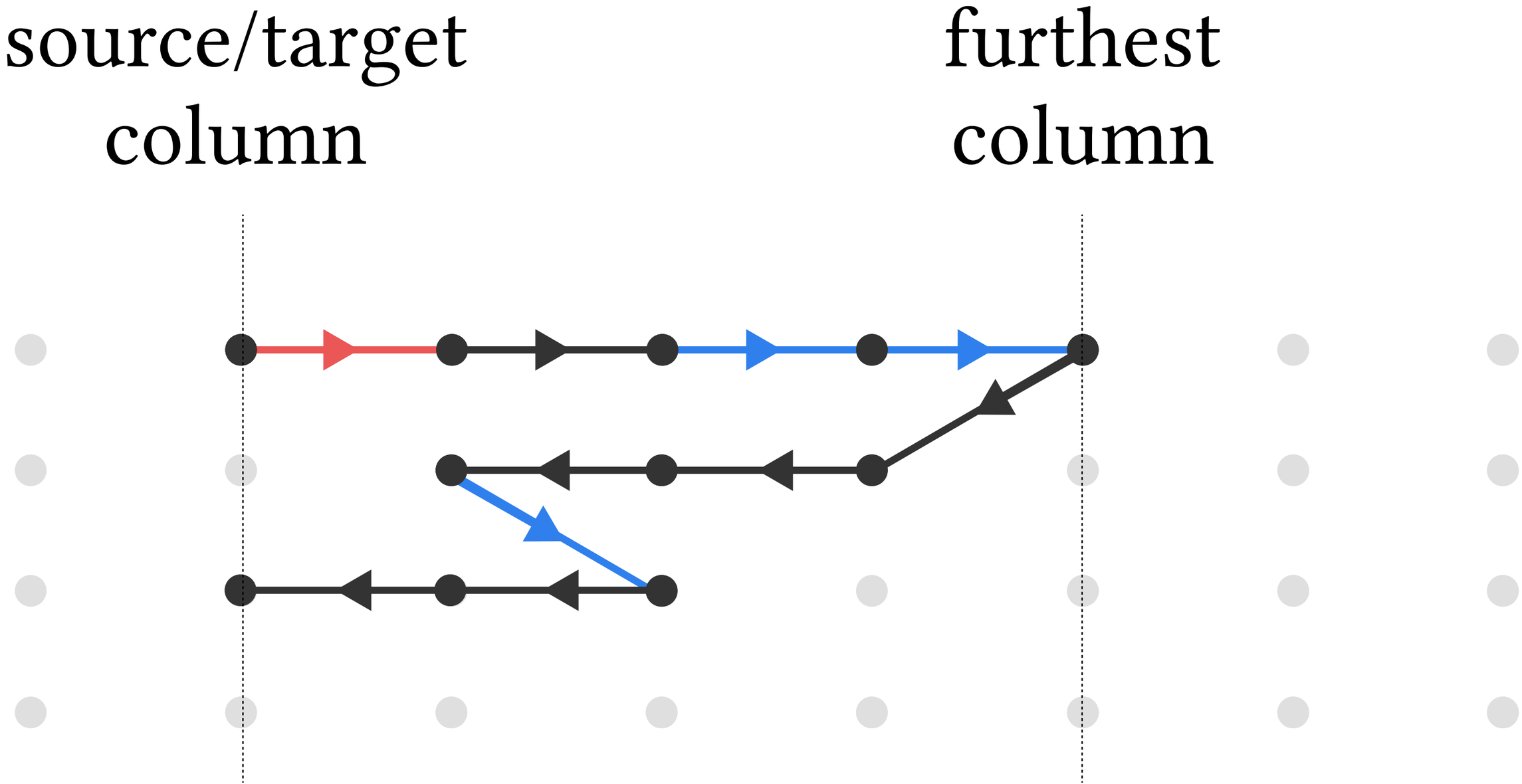

3
1
5
2
3
1
4
2
1
4
2
1
2
7

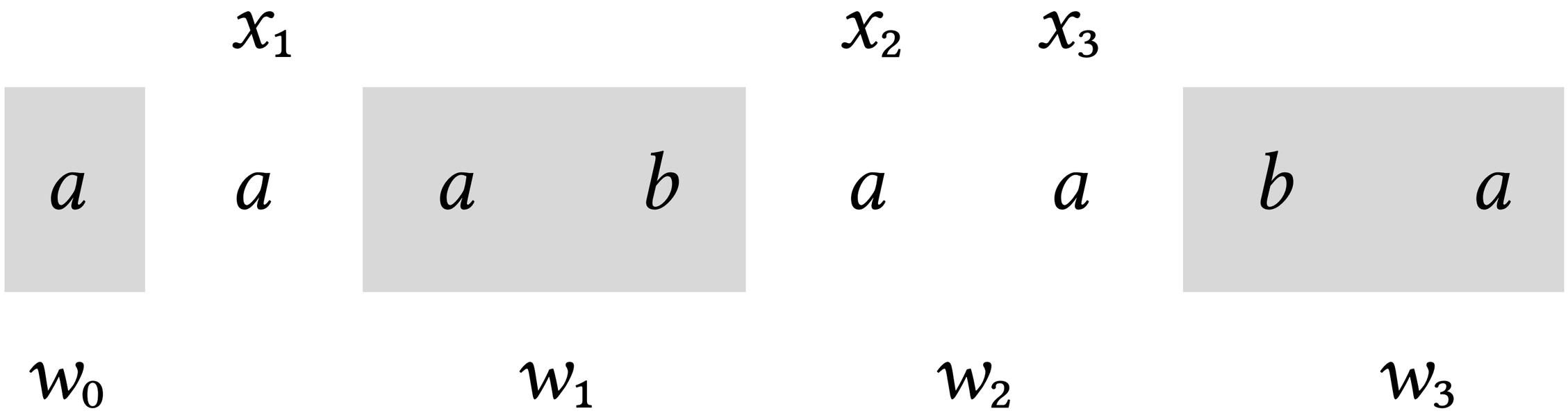


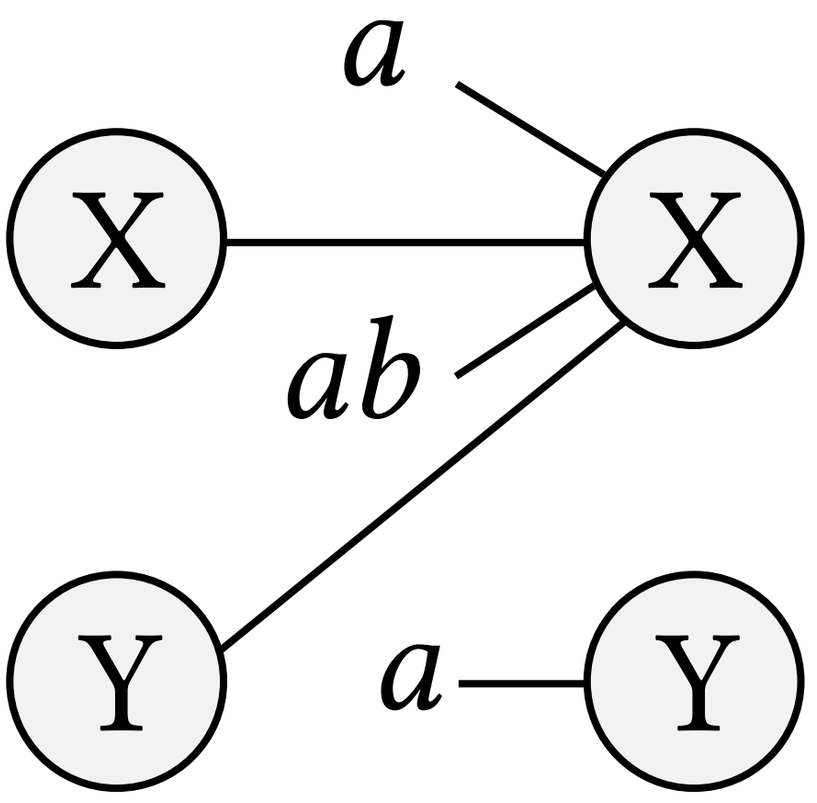

these columns are visited 3 times


x_{i-1}
x_i
x_{i+1}



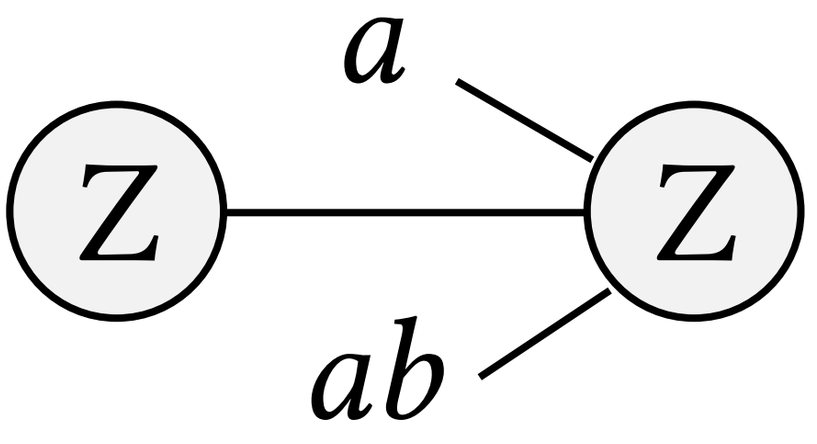

source/target
column
furthest
column


x_1
x_2
x_3
a
a
a
b
a
a
b
a
w_0
w_1
w_2
w_3

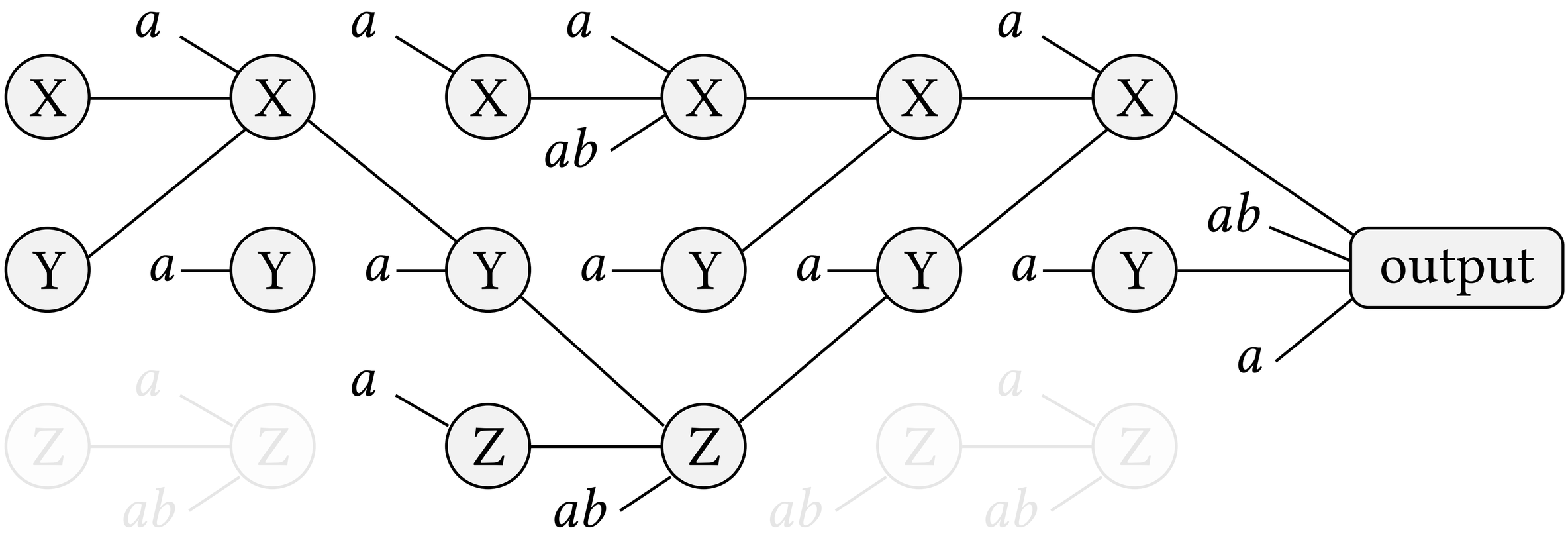


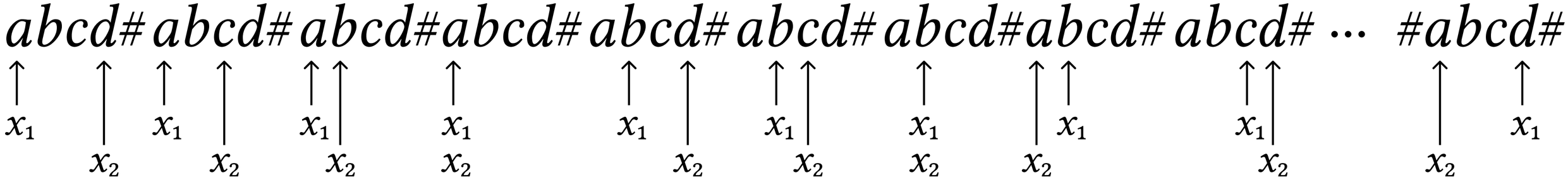
$\mathrm{X} := a\mathrm{X}ab\mathrm{Y}$

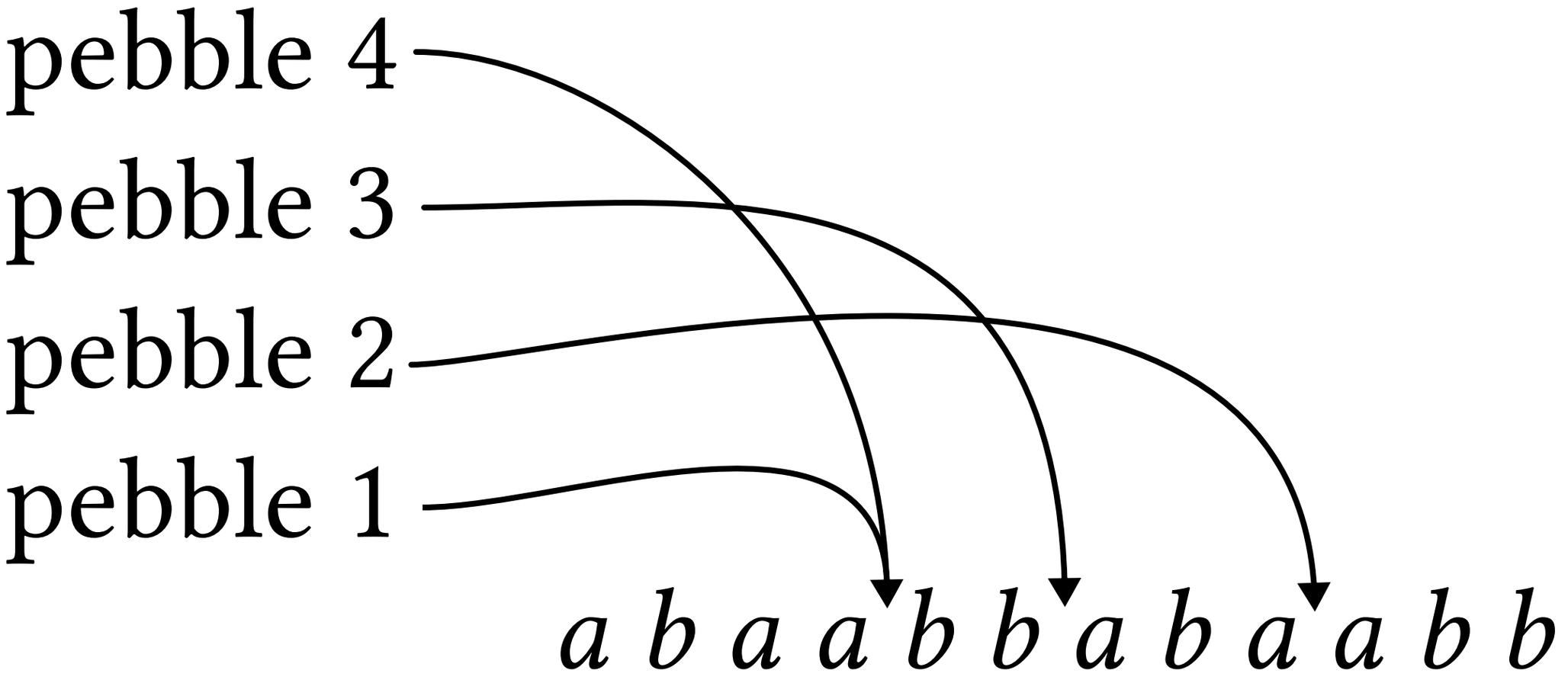
$\mathrm{Y} := a$

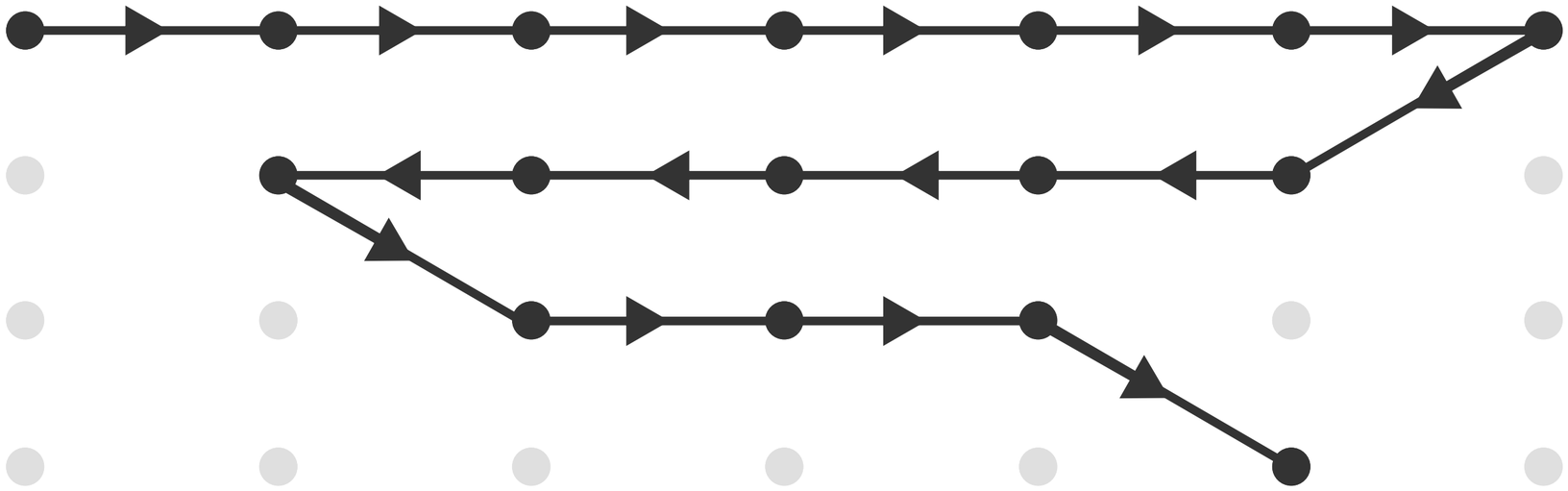


$\mathrm{Z} := a\mathrm{Z}ab$


a
a
a
a
X
X
X
X
X
ab
Y
a
Y
a
Y
a
Y
a
Y
ab
output
a
a
a
Z
Z
ab
ab
Z
Z
ab
Z
Z
ab


abcd# abcd# abcd# abcd# abcd# abcd# abcd# abcd# abcd# ... #abcd#
x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2


pebble 4
pebble 3
pebble 2
pebble 1
a b a a b b a b a a b b



movement of pebble $l+1$

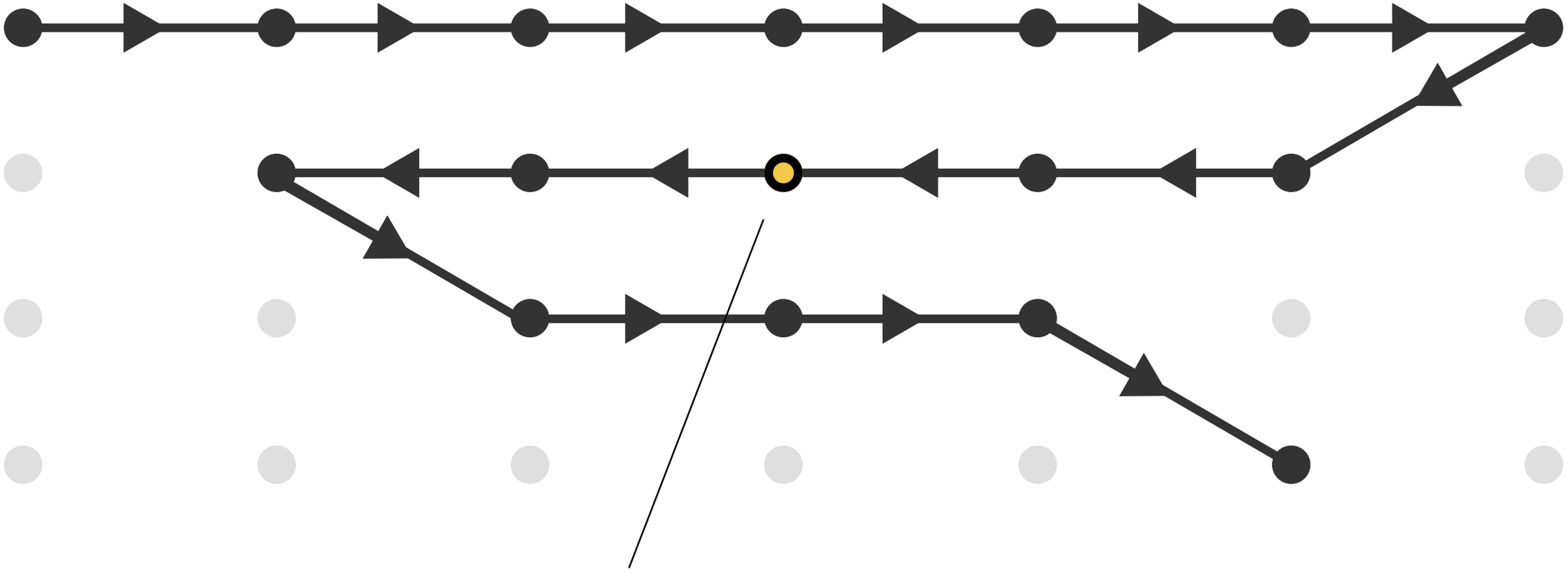
a_1 a_2 a_3 a_4 a_5 a_6

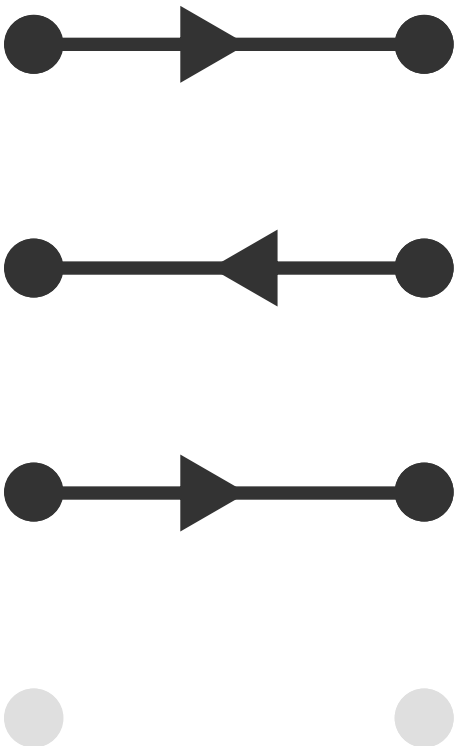
input string

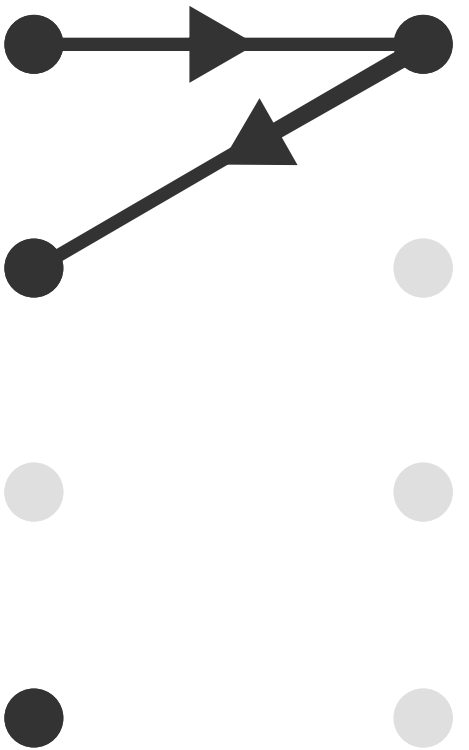
x_1 x_2

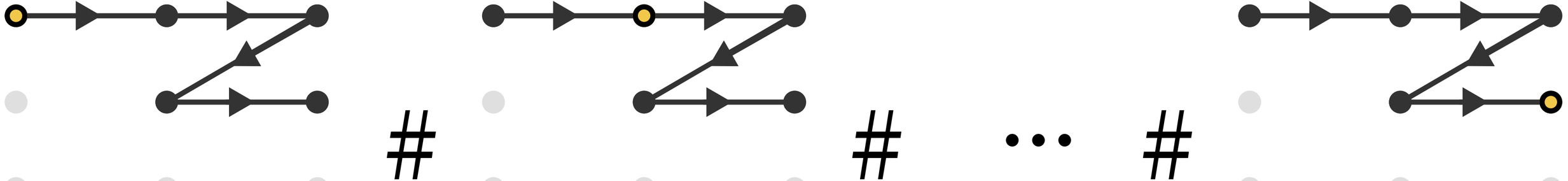
x_1
x_2

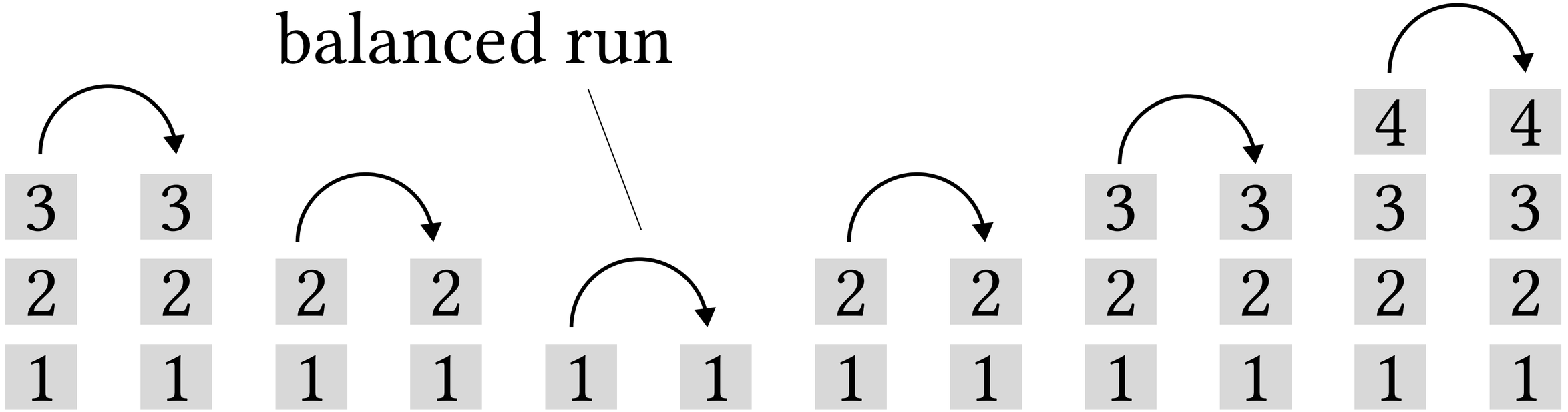
pebbles 1,...,l

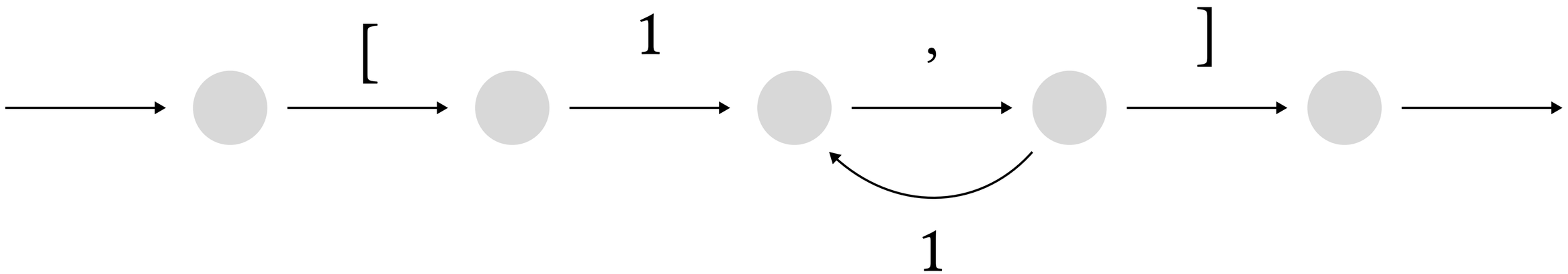

highlighted

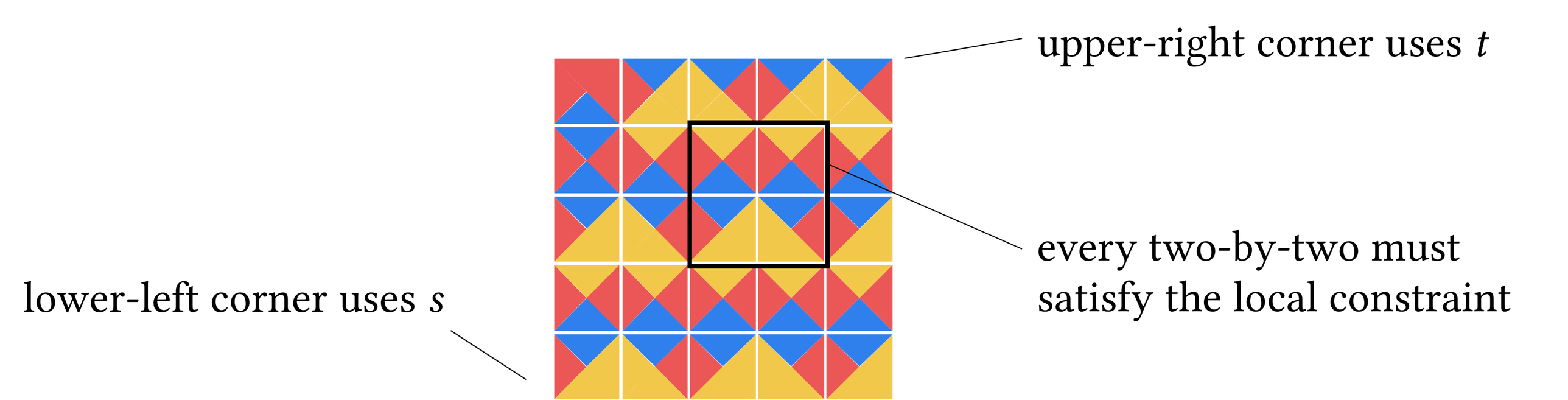







balanced run
3 3
2 2
1 1
2 2
1 1
1 1
2 2
1 1
3 3
2 2
1 1
4 4
3 3
2 2
1 1


[
1
,
]
1


upper-right corner uses t
every two-by-two must
satisfy the local constraint
lower-left corner uses s